

Hobby-level Robot Arm for Cinematography
MIE243 - Design Project

Date: Tuesday, December 2, 2025

Samit Chenchu 1011040302

Curtis Ho 1011059845

Shafwat Khan 1010976213

Soham Kundu 1011045806

Allan Zong 1011081669

Submitted To: Dr. Matthew Mackay

Executive Summary:

In the past decade, robotics has become a transformative force in the cinematography industry, reshaping how films are produced and pushing the boundaries of cinematic storytelling. Traditionally, capturing complex shots has required a mix of tools ranging from dolly tracks, cranes, and even helicopters, and the advent of robotic arms has helped introduce a new dimension that simplifies filming and creates unparalleled flexibility. Modern robotic arms often work much like Computer Numerical Control (CNC) machines, taking in a G-code file that instructs the robotic arm on the movements it has to achieve. The purpose of this report is to detail the research conducted and explain the rigorous design processes our team has pursued to achieve the final design of our very own hobby-level robot arm for cinematography. Detailed in this document is research our group has done, including benchmarks of existing designs on the market. The team has created an engineering specification based on these designs and the specific project guidelines. Most importantly, the objective is to design for a hobby-level user, and hence the design should be lightweight and affordable, yet still competitive with more advanced, professional designs.

Upon detailing the engineering specifications, our team brainstormed four candidate designs. The designs generally consist of three main components: a base with two rotational degrees of freedom around the Z and Y axes, a long arm that linearly actuates along its length depending on the base angle, and an end arm with rotational joints allowing movement around the Y axis and along the arm's axis. Together, these components enable flexible positioning and orientation of a camera mounted on the end arm. With the four candidate designs laid out, iterations were made to each individual candidate to make improvements on the weaknesses that could be mitigated.

The final design selection was performed using a pugh matrix, where each of the four main candidates were evaluated against the criteria derived from the engineering specification. Candidate design 2 was selected as the baseline and ultimately became our final design. The report details further component analyses for the final design, which also includes technical drawings and a cost breakdown. Project management was crucial throughout the process, so it is also highlighted.

Table of Contents

1.0 Project Requirements.....	4
1.1 Introduction.....	4
1.2 Benchmark Designs.....	4
1.3 Engineering Specifications.....	8
2.0 Candidate Designs.....	9
2.1 Candidate Design 1.....	9
2.2 Candidate Design 2.....	11
2.3 Candidate Design 3.....	13
2.4 Candidate Design 4.....	14
3.0 Final Design Selection.....	16
3.0.1 Pugh Matrix Evaluation Criteria and Rationale.....	16
3.1 Final Design Component Analysis.....	19
3.2 Technical Drawings.....	28
3.3 Cost Analysis.....	30
4.0 Project Management.....	32
5.0 Appendices.....	33

1.0 Project Requirements

This section of the report outlines the scope of the design project by describing the problem presented and the design objectives specific to the project.

1.1 Introduction

The objective of this project is to develop a detailed conceptual design for a long-reach robotic arm intended for hobby-level cinematographers. The design will enable hobbyists and beginner cinematographers to mount a lightweight camera to the arm's end-effector in order to create smooth cinematic moving shots at a low cost. In professional cinematography, such camera movements are often created through a combination of tools such as dolly tracks, cranes, and even helicopters. These professional options are typically very expensive, heavy, and often require multiple operators, putting them out of reach for beginner cinematographers. A well-designed robotic arm offers a practical alternative that provides controlled motion without the cost or complexity of professional equipment.

This report outlines the complete engineering process used to create such a design. The work begins with an examination of the current state of the art robotic arms and continues through to the development of detailed engineering specifications, idea generation using candidate designs, and a final design selection. The whole process is iterative, and the final conceptual design is chosen based on factors that align best with the engineering design specifications and objectives. The goal is to produce a clear and realistic concept that is suitable for further development in the detailed design phase, which can be built with accessible components while supporting smooth and repeatable camera motion for beginner cinematographers.

1.2 Benchmark Designs

Conducting research into the state-of-the-art robotic arms for cinematography that are present in the market provides a solid foundation in understanding and direction when creating a new and comprehensive design with similar qualities. Our research consists of a wide variety of such designs critically analyzing each design's strengths, weaknesses, and key takeaways focussing specifically on factors such as placement of heavy motors, overall weight, arm reach, payload capabilities, arm speed, and materials selection. Among the wide variety of designs that were analyzed, common metrics among the factors mentioned previously were taken into consideration and adjusted to meet our project's objective and consolidated into a set of engineering designs. In our research, we critically analyzed designs such as the Bolt High Speed Camera Robot, Bolt Jr+, Colossus, SISU Cinema Robots C31, Universal Robots UR3e Collaborative Robot Arm, EVO motion control robot, and many more. This section provides a detailed and consolidated description of the research that was conducted and the results associated with it.



Figure 1: Picture of Bolt Robot Arm



Figure 2: Picture of Bolt Jr+ Robot Arm

As we reviewed these systems, clear patterns emerged in both their capabilities and limitations. High-end cinematic robot arms such as the Bolt and Bolt Jr+ demonstrate the performance levels that professional studios require. These robot arms can carry up to 20 kilograms and reaches speeds up to 12 meters per second. It has a reach of 2 meters and a height of 3.5 meters and offers 6 degrees of freedom. Its structure supports target tracking and the programming of keep out zones. However, the Bolt requires a track system, a large high voltage power source with three 400 volt lines at 32 amps, and a heavy base that brings the total weight to about 600 kilograms. It also needs separate accessories for independent head tilting and preprogrammed lens control. These features highlight the impressive capability of professional equipment but they also illustrate why such systems are unsuitable for a portable and affordable long reach arm intended for beginner users.

The Bolt Jr+ offers a more compact version of the Bolt while maintaining similar design goals. It carries up to 10 kilograms, reaches speeds of 11 meters per second, and has a reach of 1.6 meters with a height of 2.5 meters. It remains portable compared to the larger Bolt and it supports target tracking and keep out zones. However, it still depends on a track and it is not as strong or fast as the Bolt. Additionally, it requires accessories for head tilting and lens control and uses a single 200 to 220 volt AC power input. Although the Bolt Jr+, is smaller, it remains too complex and too demanding for a beginner or hobbyist level focussed design.

Larger robots such as the colossus and the SISU C31 reveal another dimension in cinematography robotics. The Colossus can go 8 feet (2.44 meters) under its base and carries up to 50 pounds (22.68 Kg). In contrast to both the Bolt and the Bolt Jr+, the Colossus does not require a track as it uses large wheels and offers simple joystick control. Additionally, it is extremely large at 14 feet (4.27 meters) tall with a reach of 10 feet 2 inches (3.10 meters) and a speed of 7 meters per second. Its size and weight make assembly and setup difficult and create safety concerns. The SISU C31 offers high precision with repeatability within 0.05 millimeters and can go 1.5 meters under its base. Additionally, its top speed is 9.7 meters per second and uses significant power as it is heavy and difficult to move. These designs show the advantages of long reach and high stiffness but they also demonstrate that such capabilities require large structures that are not best suited for beginners or hobby level use.



Figure 3: Picture of EVO Motion Control Arm



Figure 4: Picture of Universal Robots UR3e

This indicates that more compact robots with slightly reduced capabilities offer qualities that are more relevant to this project and our objectives. The Universal Robots UR3e is small, easy to move, and requires low power at a maximum of 300 W consumption. It provides 6 degrees of freedom and has a reach of 500 millimeters (0.5 meters) and can carry a max payload of 3 kilograms. These traits make it accessible and simple to operate. However, the short reach and low payload capability limits its ability to create dynamic cinematic movement. The EVO Motion Control Robot provides a more balanced approach. It is compact, portable, and designed for creators who need flexibility. It has a reach of about 1 meter and supports payloads between 5 and 7 kilograms which is suitable for mirrorless and compact cinema cameras. It also has 6 degrees of freedom, brushless servo motors with encoders, and a tripod compatible base with quick release mounting. Additionally, it operates on 24 volt DC power or through an AC adapter

and supports an optional battery pack. Although the EVO offers strong usability, its short reach limits its ability to produce wide sweeping shots and it cannot handle heavier cameras.

Across all the designs and models considered, the research shows clear trends that guide the practical direction for this project and our design. Long reach and high payload capacity require large and heavy motors and these increase cost and limit portability. Systems that achieve high speed or precision also rely on strong structural components that add weight and complexity. Smaller portable robots offer ease of use and low power requirements but they lack the reach need for more dynamic cinematography. A balanced design for a beginner focused or hobbyist level long reach arm will likely use lightweight materials such as carbon fiber or aluminum and efficient motor placement. Placing motors near the base and using belt or chain driven joints can reduce weight at the end of the arm. Based on our research, a practical design should aim for a reach between 1 to 2 meters with a payload range between 3 to 7 kilograms. It should remain simple to assemble and operate while avoiding unnecessary professional features that increase the cost. Overall, the research confirms that smooth and repeatable motion, structural stability, and ease of use will provide the most value for hobbyists and beginner level cinematographers.

1.3 Engineering Specifications

The following engineering specifications are derived from comprehensive market research as described in section 1.2 Benchmark Designs. The objectives aim to balance high performance with practical considerations such as cost, manufacturability, and usability. The final design must meet or exceed the following metrics to be competitive while remaining accessible to our target user base:

1. General requirements:
 - a. Should be user-friendly and simple to assemble. Assembly time **must not exceed 45 minutes.**
 - b. Should not require exotic or difficult-to-source materials or components.
 - c. Total cost **must not exceed \$3,000** to ensure affordability.
 - d. Should require **minimal maintenance** throughout its operational lifetime.
2. Performance requirements:
 - a. Should have a **minimum operating reach of 1.5 meters.**
 - b. Should support a **payload ranging between 3 to 7 kilograms.**
 - c. Should operate at a speed between **5 and 6 meters per second.**
 - d. Should provide at least **6 degrees of freedom.**
3. Physical constraints:
 - a. Must not exceed **100 kg.**
 - b. Must operate on a standard electrical power source, **compatible with both 110V and 220V** AC inputs.
4. Additional considerations:
 - a. Noise level: should aim for quiet operation
 - b. Durability: components should be robust enough to withstand operational stresses.
 - c. Safety: should comply with safety standards.

2.0 Candidate Designs

This section includes rough engineering sketches of 4 potential designs that explore different structural layouts, motion strategies, and component arrangements. Each design includes the critical dimensions needed to understand overall footprint and major components. These concepts serve as the foundation for comparison and selection. The designs begin with broad and intentionally different approaches to explore the solution space and then progress through iterative refinement. This process allows the concepts to gradually converge toward the final design that best satisfies the engineering goals and performance requirements established earlier in the report.

2.1 Candidate Design 1

This design is built around four main joints that together provide the necessary reach and motion for smooth cinematographic movements. The end effector is supported by Joint 4, which is responsible for the camera's fine control. This joint allows rotation in two axes and includes a dedicated camera mount that can pitch up and down for angled shots, while an additional rotational axis enables controlled panning. Because this part of the arm carries a low payload and must move with high precision, Joint 4 uses brushless DC motors paired with a low torque gearbox to ensure smooth and accurate motion.

Moving down the arm, Joints 2 and 3 extend the structure and guide the arm's mid-segment movements. These joints are driven by stepper motors selected for their reliability and ability to hold positions steadily during motion sequences. Their role is to provide predictable forward and upward reach while keeping the arm stable.

Joint 1 forms the base and performs 2-axis rotation that allows the arm to both rotate horizontally and adjust its height. The horizontal rotation is applied at the lowest point of the base, while the pitching mechanism is enclosed within a dome-like housing that turns with the base to maintain smooth, continuous motion. Since Joint 1 supports the majority of the arm's mass, it is equipped with a planetary gearbox that maximizes torque and ensures controlled, powerful movement.

Overall, this candidate design combines a strong and stable base, reliable mid-arm motion, and highly precise camera control at the end effector, creating a well-rounded concept suitable for further development in the design process.

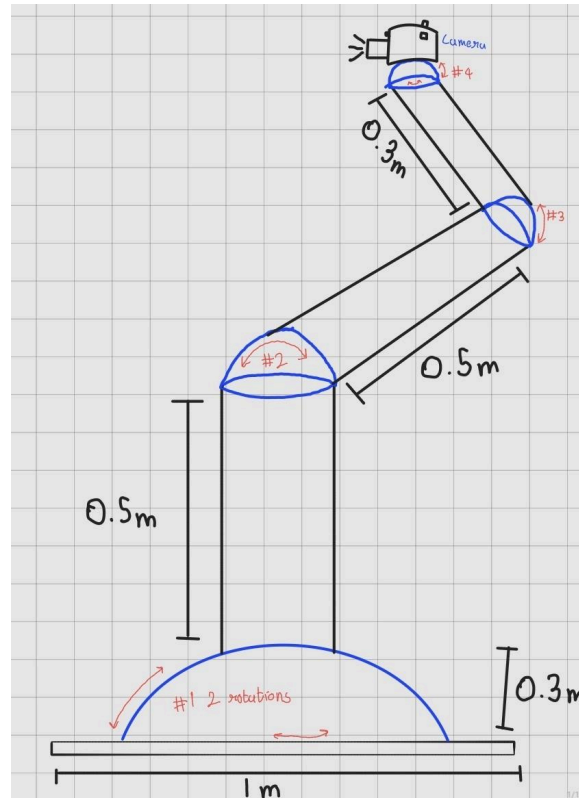


Figure 5: Rough Hand Drawn Engineering Sketch of Candidate Design #1

Throughout the development of Candidate Design #1, several key iterations were made to ensure the concept aligned more closely with the engineering design specifications. The initial approach relied on ball joints to achieve multi-axis movement, but this configuration proved difficult to implement reliably. Ball joints introduced complications in manufacturability, precision, and ease of assembly, which conflicted with the requirement for a user-friendly design that could be assembled in under 45 minutes with accessible components. Replacing the ball joints with multiple single-axis joints not only simplified construction but also improved control accuracy, which supports the requirement for smooth movement and durable operation. A second major iteration involved refining the base joint. The base was redesigned to include a planetary gearbox at the lowest point to provide strong and controlled rotation about the Z axis, while an additional joint above it handles rotation in a different plane. This change increases torque capacity at the base where the load is greatest and ensures stable support for a 3 to 7 kilogram payload. It also improves reach, stability, and precision during motion, which helps the design meet the 1.5 meter reach requirement and the target speed range of 5 to 6 meters per second. Together, these iterations improved reliability, manufacturability, cost feasibility, and overall mechanical performance, making the evolving design better suited to meet the full set of engineering specifications.

2.2 Candidate Design 2

Candidate Design #2 is built around a stable 0.30 meter-wide base that supports the arm's full six-joint configuration. The arm begins with Joint 1, which provides horizontal rotational movement and is powered by a high-torque stepper motor to ensure precise and steady base positioning. Above this, Joint 2 functions as the main load-bearing joint and incorporates a planetary gearbox to increase torque. A stepper motor drives this joint as well as Joint 3, which is mechanically linked through a belt system. This linkage allows a single motor to control coordinated motion between the two joints, reducing both weight and mechanical complexity. Joints 2, 3, and 4 operate within the same vertical plane, forming the primary lifting and extending structure of the arm.

Further up the assembly, Joint 4 introduces additional vertical-plane motion and is driven by a brushless DC motor to produce smooth and precise movement. Joint 5 adds rotation in a horizontal plane that is perpendicular to the motion of the mid-arm joints, allowing the wrist section of the arm to reposition itself independently of the base. Joint 6 provides the final degree of rotational freedom and is oriented perpendicular to all previous axes. This joint is also powered by a brushless DC motor and enables full three-dimensional orientation of the end-effector. A camera is mounted near Joint 6, allowing the operator to capture diverse and flexible shot angles with stable motion control.

Overall, this design balances torque, precision, and mechanical efficiency through the combination of stepper motors, brushless DC motors, and a strategically placed planetary gearbox and belt drive mechanism, resulting in a versatile and well-controlled robotic arm suited for beginner level cinematic applications.

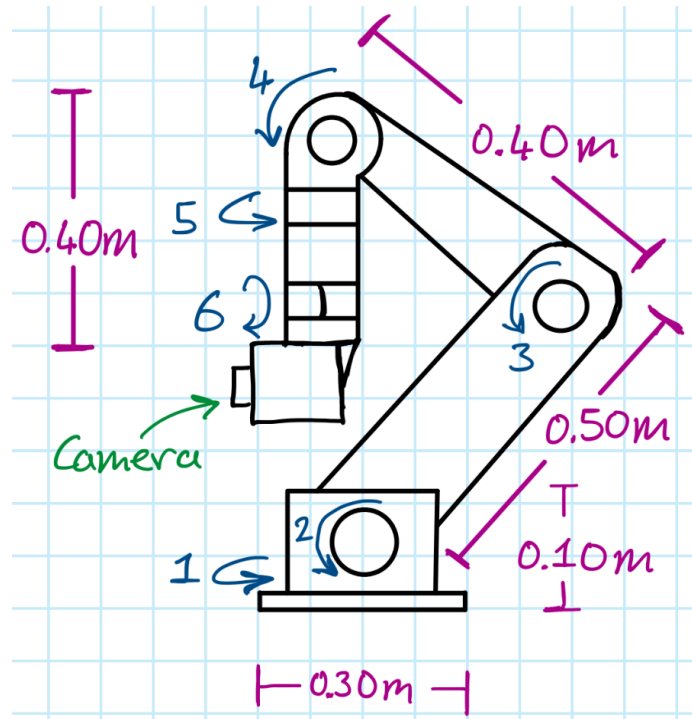


Figure 6: Rough Hand Drawn Engineering Sketch of Candidate Design #2

The design for Candidate Design #2 progressed through several iterations that improved its ability to meet the engineering design specifications. One major refinement was the addition of a linear actuator to extend the arm's reach. This modification increased the usable range beyond what the revolving joints alone could provide, helping the design move closer to the required 1.5 meter operating reach. It also improved the arm's ability to perform controlled close-up shots, which supports smoother cinematography and enhances user functionality without increasing mechanical complexity.

Another key improvement involved adjusting the belt-drive mechanism between Joints 2 and 3. In the original concept, a single motor at Joint 2 powered both joints, but this limited the independence of their motion. Separating the drive systems allowed Joint 3 to move without forcing Joint 2 to respond, resulting in smoother control and greater precision. This change increased system reliability and better aligned the design with the need for accurate and repeatable motion under a 3 to 7 kilogram payload.

The final major iteration introduced a planetary gearbox at Joint 1. Since the base joint must rotate the entire mass of the arm and payload, higher torque was necessary for safe and stable operation. Integrating the gearbox increased torque output while keeping the structure compact. This improvement strengthened the design's ability to meet durability, safety, and load-handling requirements while maintaining compatibility with standard power sources. Together, these iterations helped the design converge toward a more capable and specification-aligned robotic arm.

2.3 Candidate Design 3

Candidate Design #3 is built around the structure of a traditional camera jib, prioritizing affordability, reach, and practical use over complex robotic motion. The design uses a counterweight system with sandbags or weights at the opposite end of the arm. This balancing reduces the torque required at the major joints, allowing the use of cheaper and more accessible motors without sacrificing stability. Because the arm remains balanced during motion, Joint 2 does not need to generate high torque to keep the system in position.

The motor configuration supports this simplified but effective structure. Joint 1 uses a stepper motor paired with a planetary gearbox, providing the high torque needed to support and rotate the entire arm assembly. Joint 2 is driven by a brushless DC motor connected through a belt-drive mechanism, which shifts motor mass toward the base. This reduces vibrations and increases overall stability during operation. Joints 3 and 4 rely on a rack-and-pinion system for controlled extension and retraction, giving the arm adjustable reach with smooth linear movement. At the end-effector, Joint 5 incorporates a brushless DC motor that keeps the gimbal attachment level during operation.

Although the arm itself provides only five joints, the design integrates a universal gimbal mounting point that allows users to attach their own stabilizing gimbal systems. This approach eliminates the need to build additional rotational joints into the arm and recognizes that many hobbyist cinematographers already own such equipment. As a result, the system achieves the required degrees of freedom through a combination of the jib-like arm and the user's existing gimbal. While unconventional compared to standard robotic arms, this design offers a practical and budget-friendly solution that delivers extended reach and smooth camera motion without unnecessary complexity.

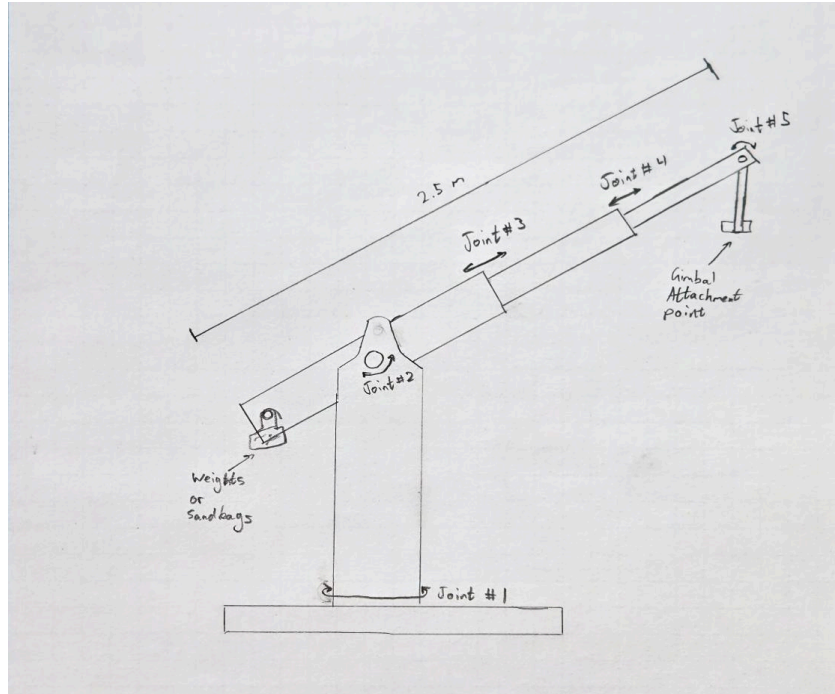


Figure 7: Rough Hand Drawn Engineering Sketch of Candidate Design #3

This candidate design underwent several key iterations to improve stability, precision, and torque handling. The original rack-and-pinion mechanism used for Joints 3 and 4 was reconsidered in favor of a linear actuator with a lead screw. This change increases stability during extension and retraction while providing more precise control of the arm's position, helping the design meet the required accuracy and repeatability for cinematic shots. Additionally, the belt-drive system at Joint 2 was enhanced with the addition of a planetary gearbox and the motor was changed from a brushless DC motor to a stepper motor. This ensures sufficient holding torque and high torque capability when the arm is fully extended and carrying the camera payload, improving stability and reliability during operation. Together, these iterations strengthen the arm's structural performance, precision, and usability while maintaining the design's focus on affordability and practicality.

2.4 Candidate Design 4

Candidate Design #4 is a seven-joint system that combines telescopic extension mechanisms with off-the-shelf gimbal technology to maximize reach and stability. The arm uses a multi-stage telescopic actuator at Joint 1, where several nested tubes of decreasing diameter slide outward. The drive motor is mounted in the largest inner tube, and motion is transmitted through a dual-belt closed-loop system. One belt extends the stages while the second belt maintains tension and stabilizes the structure, ensuring precise and controlled extension. Joints 2 and 3 provide

additional extension and retraction capabilities similar to those used in Candidate Design #3, allowing the arm to adjust its length for varied shooting scenarios.

The camera's orientation is managed by Joints 4, 5, and 6, which correspond to yaw, pitch, and roll. These three axes are integrated into a pre-existing, off-the-shelf 3-axis gimbal or camera head, providing smooth and accurate motion without the need to build complex motorized joints into the arm itself. Joint 7 functions as a ground bracing or stabilizing strut, which can be extended to make contact with the floor, a table, or other props. This addition increases overall stability, especially when the arm is fully extended. By combining telescopic reach with gimbal-based precision and external support, this design delivers optimal reach, reliable stability, and practical adaptability for a variety of cinematography setups.

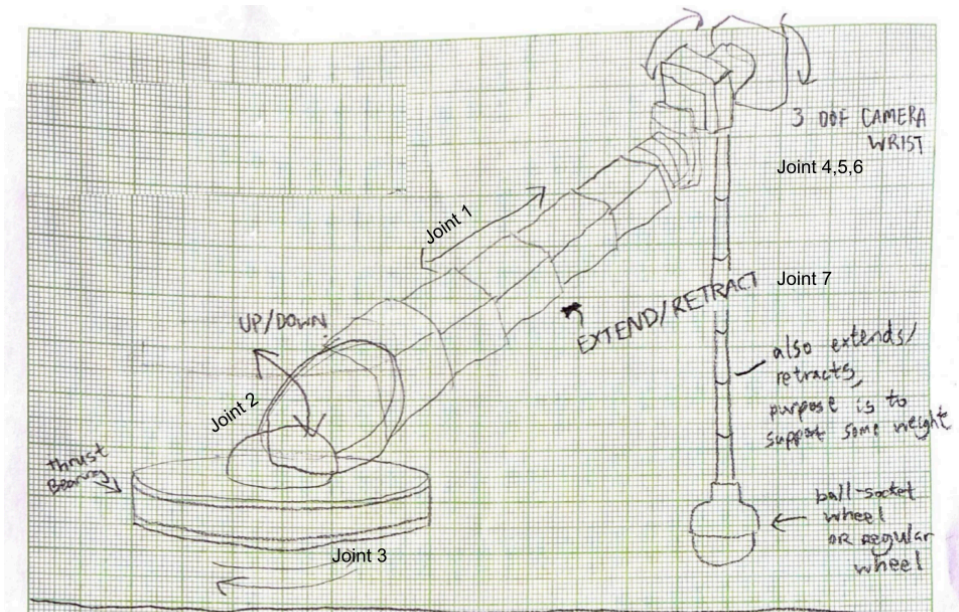


Figure 8: Rough Hand Drawn Engineering Sketch for Candidate Design #4

Candidate Design #4 underwent several key iterations to enhance stability, reach, and payload performance. The original single-legged stabilizing strut at Joint 7 was replaced with a tripod-style stand, which provides a wider base of support and significantly increases the arm's stability when fully extended. Additional reinforcement was added to Joint 2 to handle the increased torque that occurs when the telescopic arm reaches its maximum extension. This ensures the arm can safely support the intended payload without bending or failure. Finally, the original belt-drive system for the telescopic actuator was reconsidered in favor of a linear actuator with a lead screw. This change improves stability during extension, allows more precise control of the arm's position, and increases the load-bearing capacity. Together, these iterations bring the design closer to meeting the engineering specifications for reach, stability, payload, and overall reliability while maintaining practical usability for cinematography.

3.0 Final Design Selection

To systematically evaluate the three candidate designs against the engineering specifications and project objectives, a Pugh Matrix was employed. This tool provided a structured approach to compare design concepts with our outlined objectives, enabling a quantitative assessment to determine the optimal design to be implemented as the final design.

3.0.1 Pugh Matrix Evaluation Criteria and Rationale

Seven critical evaluation criteria were selected based on the engineering specifications and project requirements. The reference design (Datum) was selected as Bolt High Speed Camera Robot, which represented the most balanced approach across multiple dimensions. The scoring system employed was as follows:

+1, +2 = Better than Datum

0 = Equal to Datum

-1, -2 = Worse than Datum

Different weights, ranging from 1-3, were also assigned to each evaluation criteria based on the team's consensus of the importance of achieving the individual objectives.

Evaluation Criteria:

- **Degrees of Freedom (DOF):** Essential for achieving the required 6+ DOF. Designs with more intuitive control mechanisms and better joint architecture received higher scores.
- **Operating Reach:** Critical performance metric for achieving the minimum 1.5 m reach requirement. Designs with modular arm segments and extended reach capabilities scored higher.
- **Payload Capacity:** Required to support 3-7 kg camera payloads. Structural robustness and overall stability of the arm.
- **Motion Speed:** Designs capable of achieving 5-6 m/s end-effector speed scored higher, considering motor specifications and drive mechanisms.
- **Cost:** Must not exceed \$3000. Directly impacts manufacturability and market viability. Designs utilizing standardized components and simpler mechanisms scored favorably.
- **Assembled Weight:** Must not exceed 100 kg
- **Assembly Complexity:** Designs with fewer components, standardized interfaces, and fewer machined parts scored higher, supporting the <45 minute assembly requirement.

3.0.2 Pugh Matrix Results

The following table presents the comprehensive Pugh Matrix evaluation. Each design candidate was scored relative to the Datum (Bolt High Speed Camera Robot) across all eight criteria, yielding total scores that reflect overall design merit. The weighting of the criterias vary due to

our project's objectives. Since our design is targeted for beginner level cinematographers, cost is a major factor for the design along with degrees of freedom and assembled weight. The rationale behind the assembled weight is to keep it as light as possible, not exceeding 100 kilograms and allowing it to be as portable as possible compared to the Bolt's 600 kilogram weight. Other factors were given lower weight for the pugh chart due to trade-offs for cost and assembled weight.

Table 1: Pugh Chart evaluating each candidate design to the Bolt High Speed Camera Robot

Evaluation Criteria	Bolt Robot Arm (DATUM)	Candidate Design 1	Candidate Design 2	Candidate Design 3	Candidate Design 4	Weight
Degrees of Freedom	0	0	0	0	0	3
Operating Reach	0	0	+2	+1	0	1
Payload Capacity	0	-2	-2	-1	-2	2
Motion Speed	0	-1	-1	-2	-2	1
Total Cost	0	+1	+1	+1	+1	3
Assembled Weight	0	+1	+2	+1	+1	2
Assembly Complexity	0	0	+2	+1	-1	1
Total Score	0	0	6	3	-2	

To sum up the results of the pugh chart, it is clear that candidate design 2 aligns most closely with our project's objectives and engineering design specifications. While there are significant trade-offs due to cost and portability, candidate design 2 provides the best balance between a high quality robotic arm and a cheap ineffective robotic arm.

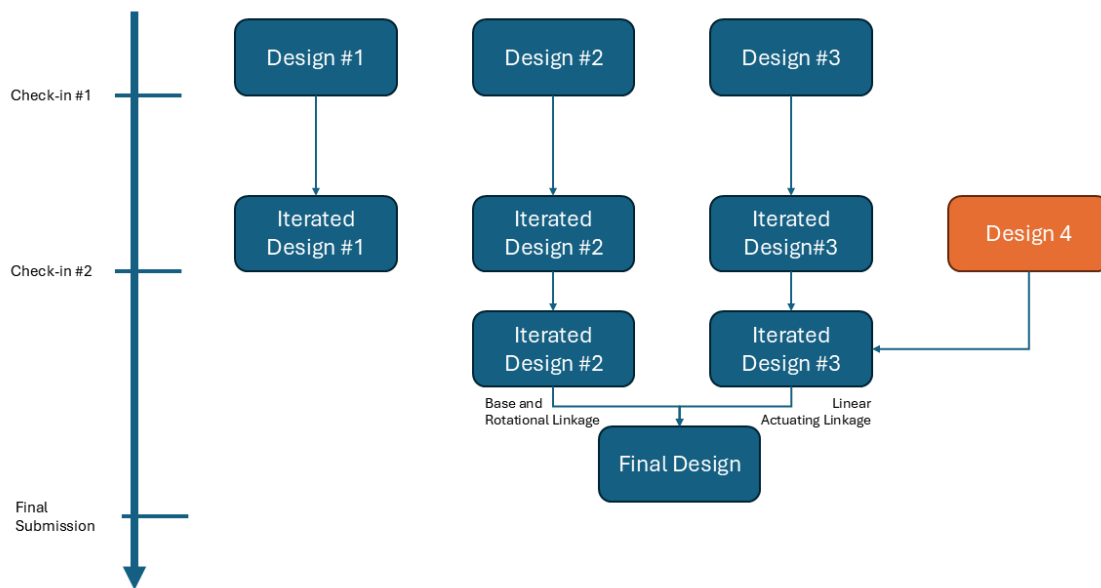


Figure 9: Iterative Evolution of Designs

Figure 9 illustrates the iterative evolution of our concepts, tracking how initial designs were evaluated and refined across the project timeline. The final candidate was selected by synthesizing the stable base and rotational linkage of Design #2 with the superior linear actuating linkage developed in the Design #3 and #4 branches.

The final design was developed by combining the most effective features from all four candidate designs and selecting the mechanisms that best meet the engineering specifications. While the overall structure follows a format similar to Candidate Design 2, the final concept integrates improvements taken from the strengths of each earlier design. The arm consists of six joints, each assigned a motion strategy and motor configuration chosen to maximize stability, precision, and usability. Joints 1 and 2 use planetary gearboxes to provide the high torque required at the base, where most of the load is concentrated. Joint 3 retains the belt-drive system but positions the motor closer to the base to reduce moving mass and improve balance. Joint 4 transitions away from a rotational joint and instead incorporates a linear actuator with a lead screw, which increases reach, improves control, and removes redundancy in the plane shared by Joints 2 and 3. Joints 5 and 6 use low-torque brushless motors to provide smooth and accurate orientation at the wrist and camera mount.

The choice to use a linear actuator at Joint 4 was influenced by the mechanisms seen in Candidate Designs 3 and 4, which both used similar extension systems to increase reach and enhance stability. This modification replaces the rotational Joint 4 from Candidate Design 2,

which previously contributed limited additional motion since it moved in the same plane as Joints 2 and 3. By integrating telescopic functionality instead, the final design offers a broader operational range and more cinematic flexibility while maintaining mechanical simplicity. Through this combination of proven mechanisms and refined joint configurations, the final design achieves a strong balance of performance, affordability, and practicality suited for hobby-level cinematography.

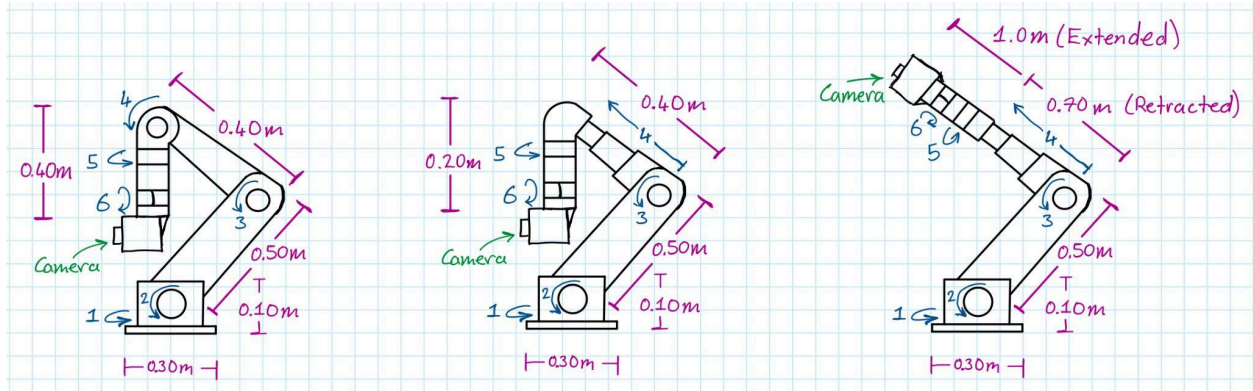


Figure 10: Iterations of Candidate design 2 to achieve the final design sketch (from left to right)

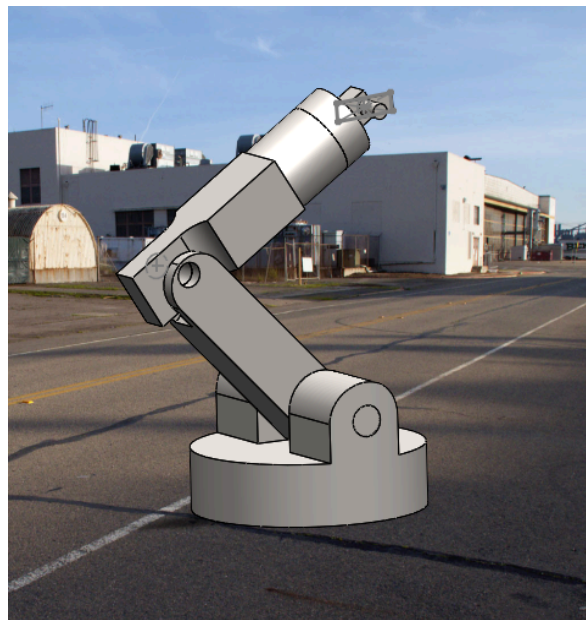


Figure 11: Initial CAD iteration of final design concept

3.1 Final Design Component Analysis

The final design is a six joint robotic arm that combines the strongest features from the earlier concepts. It is organized into three main components called the base, the long actuating arm, and the end arm that carries the camera. The base provides two degrees of freedom. It rotates about the Z axis to change the direction of the arm, which also redefines the X axis as the base turns. It

also rotates about the Y axis to move the arm through the X Z plane. Both joints use planetary gearboxes that provide high torque and low backlash. This allows the arm to move smoothly and precisely while supporting the weight of the full structure.

As seen in Figure 11, a linkage connects the base to Joint 3 and provides the rotational movement that allows the arm to fold and extend. The second linkage of the arm contains the linear actuator and the wrist joints. This linear actuator provides one degree of freedom through stable and controlled linear motion. Joint 3 uses a belt drive mechanism that is powered by a motor located near the base. The motor also uses a planetary gearbox. Positioning the motor close to the base improves stability and reduces the weight carried at the far end of the arm. This reduces bending forces and improves overall control.

The final component is the end arm. It connects to the long arm at a joint that rotates about the Y axis and allows the camera to sweep across the scene. Two additional joints at the wrist use low-torque brushless motors to rotate the camera about its axis and to control vertical pitch. These joints give the camera full control of its viewpoint without adding unnecessary weight to the end of the arm.

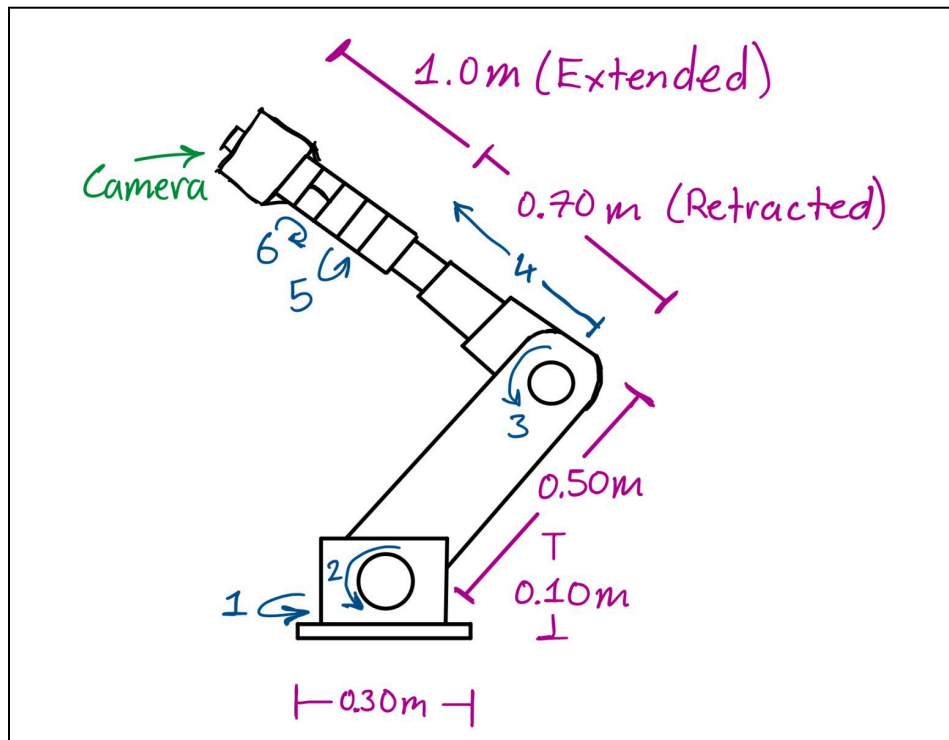


Figure 12: Rough hand-drawn engineering sketch of final design concept

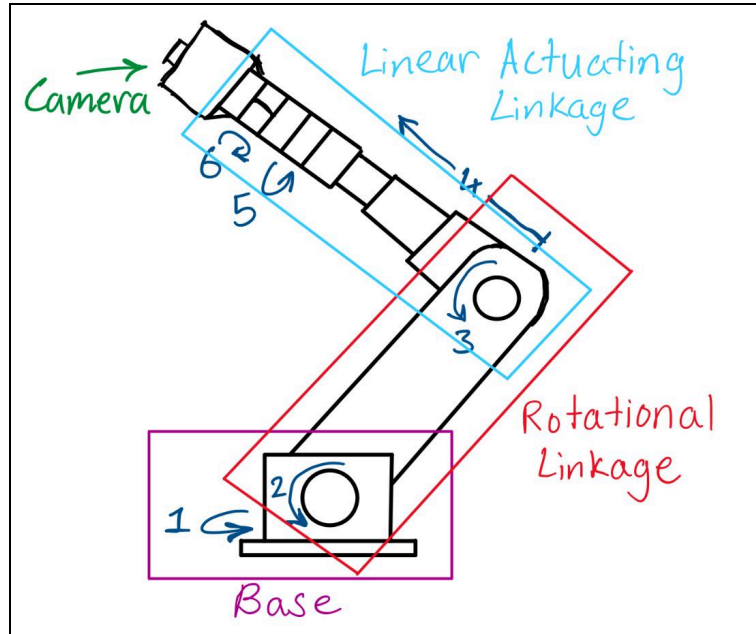


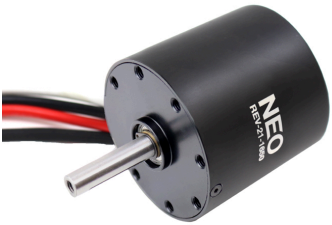
Figure 13: Reference for describing components of the arm

3.1.1 Base Components

The base is an essential part of the design because it enables rotation around the Z-axis and provides the structural support needed for the arm to move quickly and accurately. It consists of three main subcomponents: the housing and base plate, which support the full weight of the robotic arm and contain the motor and gearbox, and the two support columns, which are separate pieces that must be attached during assembly. These columns connect the rotational linkage to the base through a linkage joint. The gearbox that rotates the base is made from a series of smaller planetary gearboxes coupled together to produce a 75:1 speed reduction and torque increase. Table 2 outlines the rationale for each component and mechanism used, as well as the cost and performance trade-offs associated with them. The motor produces approximately 2.6 Nm of torque, which, when coupled with the 75:1 gearbox, results in an output torque of 195 Nm. The base weighs around 86.7 kg and the motor is responsible for rotating it. The motor itself does not directly support any of this weight. The moment of inertia about the relevant axis is approximately $33 \text{ kg}\cdot\text{m}^2$ in the worst case scenario (when the arm is laying flat). Using the formula for angular acceleration, α equals T/I , giving $\alpha = 195/32.99 \approx 5.91 \text{ rad/s}^2$. After completing a 90 degree rotation, the base will reach an angular velocity of about 2.12 rad/s , or approximately $121.5^\circ/\text{s}$.

Table 2: Description of the Trade-offs and design choices for the Base components.

Component/Mechanism	Rationale	Trade-offs
Multi-Stage 75:1 Torque	Increases the torque supplied	Planetary gearboxes cost

Increase Planetary Gearbox - 2 x REV 5:1 Planetary Cartridge - REV 3:1 Planetary Cartridge 	by the motor so the base can rotate the full arm assembly with ease [1]. Maintains low backlash which improves precision and smoothness in camera movements. Keeps the output shaft collinear with the motor shaft, which saves space and supports a compact and efficient layout [1]. Enhances load-handling capability, especially when the arm is fully extended and the rotational joint experiences higher forces.	more than simpler options such as spur gears or belt drives, but their improved torque capacity, compactness, and low backlash provide significant performance advantages [1]. A high gear ratio reduces the maximum rotational speed of the arm. However, the motor's input speed of roughly 6800 rpm is reduced to about 90.7 rpm, which still allows the arm to complete a full 360-degree rotation in under one second.
NEO Vortex Brushless Motor 	Provides high efficiency and delivers consistent power output during operation [2]. Offers a high power-to-weight ratio, which makes it suitable for driving heavy loads when paired with the 75:1 planetary gearbox [2]. Uses brushless motor technology and includes built-in encoders that supply accurate position feedback for precise movement control [2].	More expensive than stepper or brushed motors, but the added cost is justified because built-in position feedback is essential for achieving accurate camera coordinates. Difficult to access and replace due to its placement within the design, which can increase maintenance time and overall service costs.

3.1.2 Rotational Linkage Components

The rotational linkage can be broken down into 2 joints, Joint 2 and Joint 3 as seen in Figure 13.

Joint 2 consists of a high-torque stepper motor (WO-8718L-08P-RO), a 50:1 gearbox, and the arm itself. The stepper motor can output approximately 9 N·m of torque, which is amplified to 450 N·m after the gearbox reduction. The arm is composed of three separate parts. This design

allows the linkage rods of Joint 3 to be inserted into the slots of Joint 2 first, after which the components can be secured onto the arm of Joint 2. This modular approach makes assembly and disassembly straightforward.

The arm connects to the supporting pillars of the base via a linkage joint. It can first be attached to the linkage rods of Joint 2 and then fastened securely to the base. Based on SolidWorks mass properties, in the extreme case when the arm is fully horizontal, the motor in Joint 2 is responsible for moving approximately 36.9 kg of mass, with a perpendicular distance of 0.879 m from the axis of rotation.

Assuming 90% efficiency for the gearbox, the total torque output at the joint is 405 N·m. The torque applied by the arm due to gravity is approximately 342 N·m, resulting in a net torque of roughly 63 N·m. Using the rotational dynamics equation: $T_{net} = I \cdot \alpha$, where I is the moment of inertia about the motor axis (approximately 83 kg·m² from SolidWorks), solving for angular acceleration gives: $63/83 \approx 0.75 \text{ rad/s}^2 \approx 43^\circ/\text{s}^2$. This means that the arm would reach a speed of approximately 88.9°/s after rotating 90°.

Joint 3 can be broken down into 3 subcomponents. There is the REV NEO Brushless motor which is the same one used in the base to rotate the entire arm assembly. This motor is coupled with a multi-stage planetary gearbox that provides a 25:1 torque increase allowing the efficient movement of the end arm components. As the gearbox and the motor increase weight and destabilizes the arm if they are mounted further from the base, the design employs a 1:1 gear ratio belt drive mechanism to allow for the motor and the gearbox to be mounted near the base and the belt drive is able to simply transmit the torque generated to the end of this linkage and able to rotate the last major component of the arm which is the linearly actuating component. In Table 3, the descriptions of the specific components are provided along with rationale for choosing these specific ones. Additionally, performance and cost tradeoffs are also discussed to justify our design choices and components selections.

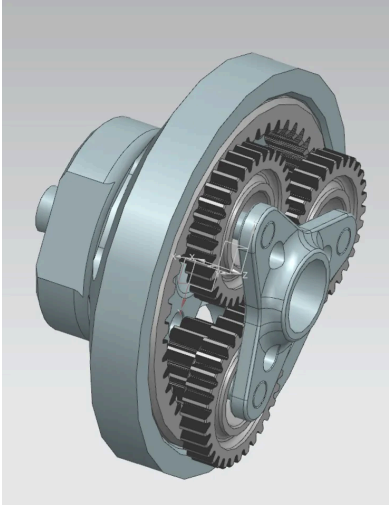
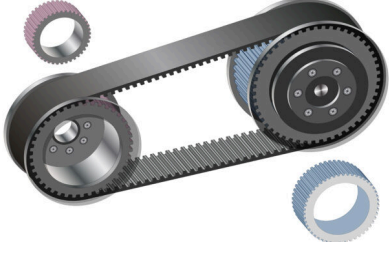
To justify our component decisions, the math for joint 3 is shown here. The angular acceleration with the minimum requirement of 90° of angle for the end effector to reach 6 m/s.

$\alpha = \frac{(6 \text{ m/s})^2}{\pi} = 11.56 \text{ rad/s}^2$. Which can then be inputted into the torque formulas, which includes the mass of everything the joint is rotating of 23kg and the center of mass of 0.173m, and finally accounting for gravity, the final necessary torque is:

$$T_{net} = (23 \text{ kg} \times 9.81 \text{ m/s}^2 \times 0.173 \text{ m}) + (23 \text{ kg} \times (0.173 \text{ m})^2 \times 11.56 \text{ rad/s}^2) = 46 \text{ Nm}.$$

With this, we chose our components to provide the overhead of around 20Nm of torque for higher acceleration or higher speed uses than the minimum design specifications require.

Table 3: Description of the Trade-offs and design choices for the Rotational Linkage Components.

Component/Mechanism	Rationale	Trade-offs
Multi-Stage 25:1 Torque Increase Planetary Gearbox REV 5:1 Planetary Cartridge REV 5:1 Planetary Cartridge 	Enables faster movement of the arm and linkage. Requires less torque than the base motor since it moves a smaller mass. Maintains low backlash, which helps prevent jerky movements and ensures smooth camera motion. Provides sufficient torque to support the arm and maintain stability even when fully extended [1].	Planetary gearboxes are more expensive than simpler options such as spur gears or belt drives, but they provide higher torque, compact design, and low backlash for precise and reliable operation [1]. A high gear ratio reduces the maximum rotational speed of the arm. With a 25:1 gearbox, the motor's input speed of approximately 6800 rpm is reduced to about 272 rpm, which still allows the arm to rotate a full 360 degrees in under one second.
1:1 Ratio Belt Drive Mechanism 4x HTD 8M Aluminum Pulley 	Enables the motor to be mounted closer to the base, improving stability and reducing load on the motor while minimizing wear and tear [3]. Absorbs shocks and vibrations, helping to protect both the motor and the arm's joints [3].	Requires careful tensioning and alignment, which increases assembly time and adds complexity [3]. Belts may stretch over time, potentially reducing precision and necessitating periodic maintenance [3].
NEO Vortex Brushless Motor	Serves the same general purpose as the base motor by providing the power and accuracy needed for controlled arm movement. <u>Additionally:</u> Uses a lower gear reduction ratio, which allows the arm to	More expensive than stepper or brushed motors, but the added cost is justified because built-in position feedback is essential for achieving accurate camera coordinates. Difficult to access and replace due to its placement within

	extend and retract more quickly than the base, resulting in faster end-effector and camera motion. Uses brushless motor technology and includes built-in encoders that supply accurate position feedback for precise movement control.	the design, which can increase maintenance time and overall service costs.
---	--	---

3.1.3 Linear Actuating Linkage Components

The linear actuation system of the robotic arm consists of a Lin stepper motor coupled to a 9:1 reduction planetary gearbox. This combination provides high torque with low backlash, allowing the motor to reliably drive the arm's linear motion. The gearbox output is connected to a high-torque timing belt system, which simultaneously drives two precision lead screws mounted along the arm. By turning both lead screws at the same time, the timing belt ensures smooth, synchronized linear motion, allowing the extending arm to move straight and accurately while supporting a payload. This configuration also keeps the motor and gearbox closer to the base, reducing the mass at the far end of the arm and improving overall stability while minimizing bending forces. The arm's end effector is mounted on joints 5 and 6, each of which has low-torque brushless motors attached directly. These motors provide wrist rotation, allowing the camera to move fluidly in multiple directions and achieve the precise orientation needed for cinematography. Together, these components create a stable, high-precision actuation system capable of both linear extension and versatile camera movement. Additionally, performance and cost tradeoffs are also discussed in Table 4 to justify our design choices and components selections.

Below are the basic calculations and values done to calculate the torque requirement for the linear actuator:

- The lead screw lead is 5 mm per screw revolution.
- The motor is coupled to the screw through a 9:1 reduction planetary gearbox. The gearbox efficiency is 0.90.
- The timing belt and pulley system efficiency is 0.98.
- The lead screw is a precision trapezoidal screw with an effective mechanical efficiency of 0.30 when converting shaft torque to linear force.
- The load is a 23 kg mass being lifted vertically. Gravity is $g = 9.81 \text{ m/s}^2$.
- All torques are in N·m and distances in meters.

The linear force required to hold or lift the mass is: $F = mg = 23 \times 9.81 = 225.63 \text{ N}$

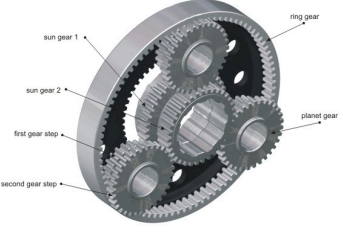
For an ideal (frictionless) screw the torque required at the screw shaft to produce linear force F is:

$$T_{\text{screw}} = \frac{225.63 \times 0.005}{2\pi \times 0.30} = 0.598 \text{ Nm}$$

Which is achieved after the 9:1 speed reduction of the motor.

Table 4: Description of the Trade-offs and design choices for the Linearly Actuating Linkage Components.

Component/Mechanism	Rationale	Trade-offs
HTD 8M Aluminum Pulley & High-Torque Timing Belt 	Used to synchronize both lead screws so the linear actuation remains coordinated and consistent throughout the full extension of the arm [4]. Capable of transmitting high torque with minimal backlash, which increases stability and ensures smooth and precise linear motion under load [4]. Enables the motor to drive both lead screws at the same time through a shared belt path, ensuring equal and straight-line extension without twisting or misalignment [4].	More expensive than low-cost imported linear actuators, but provides significantly higher accuracy, smoother motion, and more reliable performance for precise camera positioning. Offers less inherent protection than a fully enclosed linear actuator system, which may reduce overall stability if not properly supported or shielded.
Multistage 9:1 Planetary Gearbox 2x REV 3:1 Planetary Cartridge	Increases torque output from the stepper motor, allowing the lead screws to move the arm smoothly even under high load or when the arm is fully extended [1]. Provides low backlash compared to simpler gear systems, which improves	More expensive than basic spur gear or belt-only reductions due to added manufacturing complexity, precision tolerances, and multi-stage gearing [1]. Higher internal friction and gear meshing losses reduce output speed and overall

	positional accuracy and prevents unwanted oscillations or drift during linear motion [1]. Compact and coaxial design keeps the motor and gearbox aligned with the pulley system, simplifying integration and helping maintain a slim and efficient actuator layout [1].	system efficiency, resulting in slower linear travel for the same motor speed. Increased weight and mechanical complexity may require stronger mounting hardware, more precise alignment during assembly, and periodic maintenance to sustain long-term performance.
McMaster Carr 300 mm Precision Lead Screws 	Provide smooth and repeatable linear motion with high dimensional accuracy, which is essential for achieving precise camera positioning along the arm [5]. Offer high load-carrying capability, allowing the arm to extend and retract reliably even when supporting the weight of the end effector and camera [5]. The precision machining minimizes backlash and axial play, improving stability and ensuring consistent movement when paired with the timing belt and gearbox-driven system [5].	More expensive than standard threaded rods due to tighter tolerances, improved surface finish, and higher-grade materials used in manufacturing. Increased friction compared to ball screws reduces mechanical efficiency, requiring more torque from the motor and gearbox to achieve the same linear force. Lead screws may experience wear over time, especially under continuous or heavy loads, which can gradually reduce accuracy and require periodic replacement or lubrication [5].
Lin Engineering Stepper Motor	Provides high holding torque and precise incremental movement, which is essential for stable and accurate linear actuation along the lead screws [6]. The 9:1 reduction increases torque output and reduces reflected load on the motor, allowing the linear actuator to	Higher cost compared to simpler brushed motors due to the added gearbox, increased torque capability, and improved motion precision [6]. Reduced maximum linear speed as a result of the 9:1 gear reduction, which limits fast extension and retraction

	move the arm smoothly and handle high loads without stalling. Integrates well with timing belt and pulley systems, enabling synchronized rotation of both lead screws and ensuring consistent and predictable linear travel.	in exchange for higher torque and accuracy. Gearbox and stepper assembly increase system weight and add mechanical complexity, which may require more careful alignment, maintenance, and calibration [6].
---	--	--

3.2 Technical Drawings

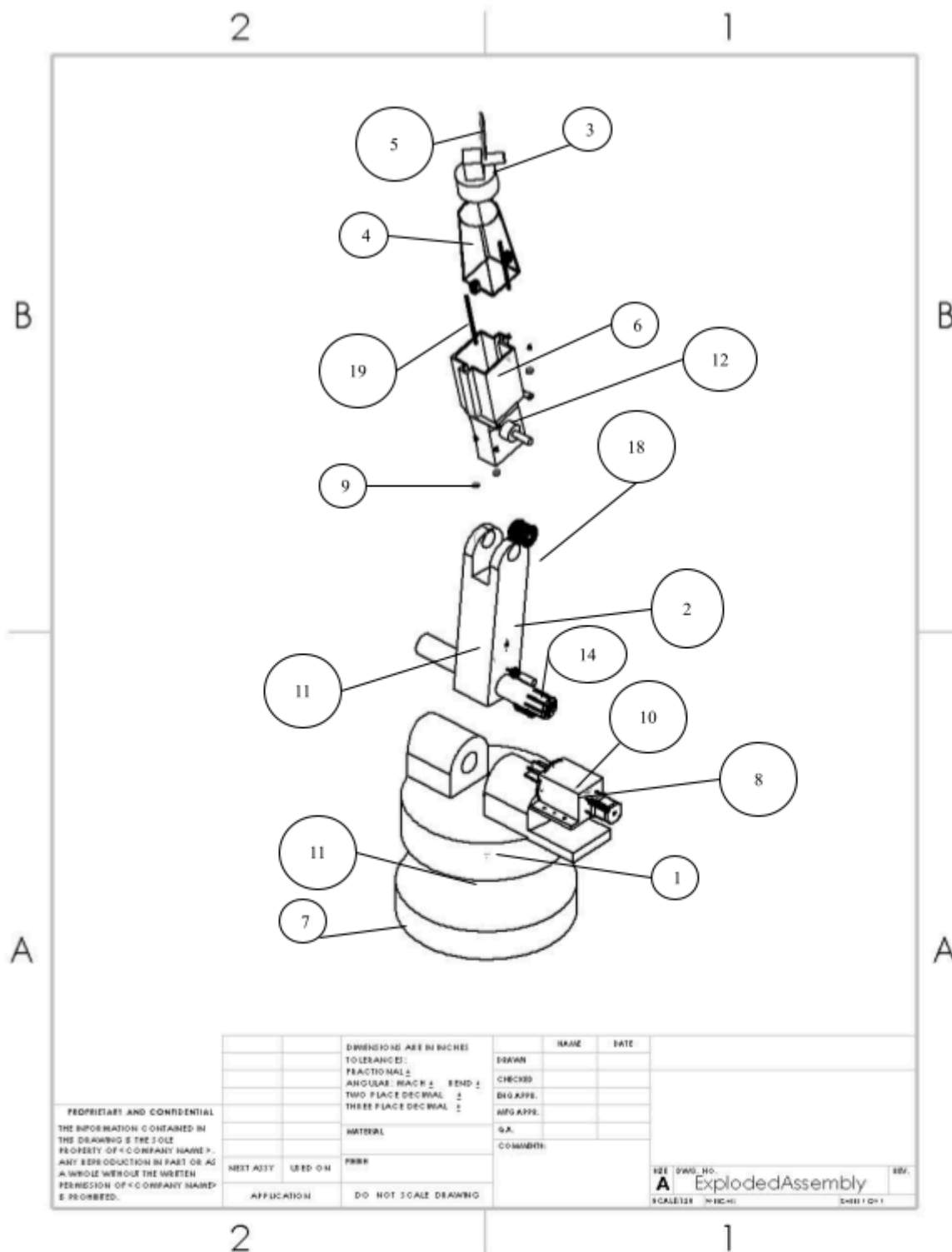


Figure 14. Exploded View with Labels

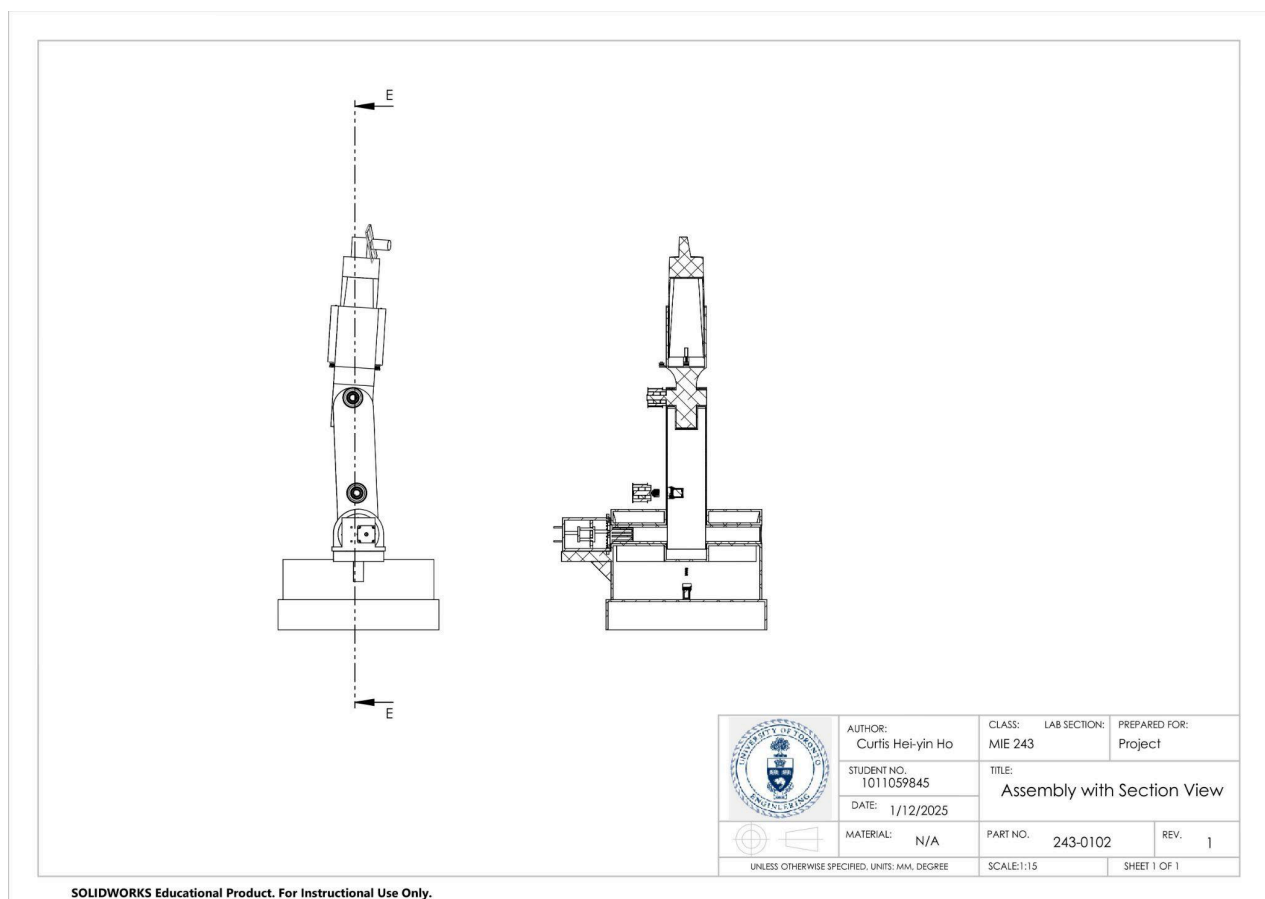


Figure 15. Cross Section View

3.3 Cost Analysis

The following cost analysis uses the cost per unit mass of Aluminum as \$2.465 CAD/kg from the Cambridge Engineering Selector [7]. We also consider a nominal labour cost of 25 CAD/hr [8]. The cost of other components are directly taken from the sites that the components are referenced from.

Table 5: Cost Analysis of Components.

Item No.	Component Description	Part Number	Quantity	Total Cost [CAD]
1	Base	243-0101	1	123.25
2	Rotational Linkage	243-0102	1	34.51
3	0.1 Beam Cylinder for Rotation	243-0201	1	8.76
4	0.4 Beam Cylinder	243-0202	1	3.69
5	End effector	243-0205	1	0.66
6	Linear Actuating Linkage	243-0206	1	67.75
7	Part 1	243-0207	1	58.88
8	Coupler for 50 to 1	243-0211	1	0.05
9	Pulley Coupler for Lead Screw	243-0213	4	0

10	High-torque Stepper Motor (WO-8718L-08P-RO)	243-0801	1	382.30
11	NEO Vortex Brushless Motor	243-0802	2	115
12	Lin Engineering Stepper Motor	243-0803	1	120
13	Sparkfun Continuous-rotation Servo:	243-0804	1	12.9
14	REV 5:1 Planetary Cartridge	243-0805	2	25
15	REV 3:1 Planetary Cartridge	243-0806	2	25
16	REV UltraPlanetary Female	243-0807	2	37
17	High-torque Timing Belt	243-0808	1	38.97
18	HTD 8M Aluminum Pulley	243-0809	1	15.03
19	McMaster 300 mm Precision Lead Screws	243-0810	2	142.1
20	100mm OD Ball Bearing	243-0811	2	35.62
21	80mm OD Ball Bearing	243-0812	2	37.2
22	Sun Gear	243-0813	3	N/A
23	Planet Gear	243-0814	9	N/A
24	38-tooth Ring Gear	243-0815	1	N/A
	Total		44	1283.67

Considering an assembly time of around 3 hours for one person to build it with raw components, there is an additional labour cost of 75 CAD, which amounts to a new total of 1397 CAD.

4.0 Project Management

In order for our team to work on such a large scale project, we implemented multiple tools and designated roles to keep the design process well ordered. The organizational tools we used were a shared google drive, shared google documents, weekly planning software, a Notion space, and a Jira board. First, a google drive was created to house all the team documents and CAD related files that would be worked on. One of the most important team documents to our organizational success recorded all meeting notes. A designated note taker maintained regularly updated notes which allowed our team to come back to earlier design ideas and to assess the development of our project. Although our class times were almost aligned, our team did not adhere to a weekly meeting period due to shifting availabilities. Instead we used the When2Meet event planner and the beginning of each week to decide what time and day would fit everyone's availability the best. Finally, our team used project boards to keep track of tasks and measure timelines. The team started off with a Notion site that listed tasks and deadlines, however when the team expanded and more tasks were undertaken in a stricter timeline greater organization was needed. Due to this our team switched to using Jira to have a greater ease of viewing ongoing tasks and even add a prioritisation ranking to our tasks (Figure 16). At the end of team meetings we would distribute tasks to each member and create these tasks on Jira with the appropriate deadlines and urgency levels.

Basic JQL Q Search work Project = RoboticArm Assignee Type Status More filters					
☐	Work	Assignee	Reporter	Priority	Status
☐	☒ ROB-6 Explain Methods of Choosing between candid...	 Samit.chenchu	 Allan Zong	Medium	IN PROGRESS
☐	☒ ROB-8 Multi-Criteria Decision Analysis	 Samit.chenchu	 Allan Zong	Medium	IN PROGRESS
☐	☒ ROB-7 Pugh Method	 Samit.chenchu	 Allan Zong	Medium	IN PROGRESS
☐	☒ ROB-5 Explain the iterations of the candidate designs...	 Allan Zong	 Allan Zong	Medium	IN PROGRESS
☒	☒ ROB-4 Start the Report Document and Start writing	 Soham.kundu	 Allan Zong	Medium	IN PROGRESS
☒	☒ ROB-11 Explain Candidate Designs	 Soham.kundu	 Allan Zong	Medium	IN PROGRESS
☒	☒ ROB-10 Engineering Specifications	 curt.ho	 Allan Zong	Medium	IN PROGRESS
☒	☒ ROB-9 Introduction and Research	 curt.ho	 Allan Zong	Medium	IN PROGRESS
☒	☒ ROB-3 Preliminary CAD model	 Shafwat Khan	 Allan Zong	High	DONE
☒	☒ ROB-2 Final Design Specifications	 Shafwat Khan	 Allan Zong	Highest	DONE
☒	☒ ROB-1 Final Design Sketch	 Allan Zong	 Allan Zong	Highest	DONE

Figure 16. A list of tasks assigned to group members using Jira.

5.0 Appendices

This section consists of the detailed research, idea generation, rough drafts and sketches and extra details relevant to the report.

Appendix A: Research conducted on existing state of the art designs

This appendix consists of the market research conducted for the state of the art designs evaluated to benchmark our project's design and engineering design specifications. This appendix specifically focuses on performance and general qualities of the robot arms.

Table 6: Table consisting of the research for each state of the art design evaluated.

Product Name	Strengths	Weaknesses	Key Takeaways
Bolt High Speed Camera Robot [9]	Carry up to 20kg Repeatable Portable Target tracking Can program "keep out zones"	Needs accessory for independent head tilting Needs an accessory for preprogrammed lens control Track Needed	2m reach 3.5m Height Bolt + Base 600kg 3x 400 volts, 32 amps, 50-60Hz 12m/s 6 Axis of Rotation
Bolt Jr+ Compact high-speed rig with extended reach	Portable Carry up to 10kg Smaller sized Target tracking Portable Repeatable Can program "keep out zones"	Not as strong or fast Lower reach Needs accessory for independent head tilting Needs an accessory for preprogrammed lens control Track needed	1.6m reach 11m/s 2.5m height 1 x 200-220 VAC
Colossus	Can go 8ft UNDER its base Repeatable Precision Carry up to 50lbs No track needed, has wheels Simple Joystick control	Set up and moving is complex HUGE takes up a lot of space Expensive Safety considerations, heavy and slow	HUGE 14 ft tall 10ft 2in reach 7m/s
SISU Cinema Robots	Repeatable with	Heavy	9.7 m/s

C31	0.05mm of uncertainty Safety switches Speed options for safety Can go 1.5m under its base	Not easy to transport Takes up a lot of space Draws a lot of power	Wing span of 20ft 14 ft tall 20kg carry
Universal Robots UR3e Collaborative Robot Arm [10]	Small Very easy to move Draws less power	Small and not as strong	6 Axis of Rotation 3kg max load 500mm reach Weight 11.2 kg Repeatability 0.03mm Max average power consumption: 300W Typical average power consumption: 100 W
EVO Motion Control Robot [11]	Compact & Portable: Designed for creators on the move; lightweight and customizable. Marketed as their most budget-friendly system. Comes with integrated software and hardware for plug-and-play operation. Suitable for tabletop, product, and dynamic	Limited Payload: Optimized for smaller camera setups. Short Reach: Prioritizes portability over extended arm length. Less Suitable for Large-Scale Shoots: Not ideal for heavy rigs or expansive sets.	Design for modular portability with foldable joints. Use lightweight materials like carbon fiber or aluminum. Include quick-connect interfaces for fast setup. Target payload of 5–7 kg for mirrorless and compact cinema cameras.

	camera shots.		Specs: 6-axis Degree of freedom Reach: ~1 meter Payload: ~5–7 kg (mirrorless and compact cinema cameras) Mounting: Tripod-compatible base with quick-release plates Motors: Brushless servo motors with integrated encoders Power Supply: 24V DC or AC adapter; optional battery pack
--	---------------	--	--

Appendix B: Research on motors and mechanisms for the candidate and final designs

This appendix outlines the research conducted on motors and mechanisms for specific locations of the robotic arm. It explains the advantages and disadvantages of each mechanism or motor, where it may be implemented and why. This research made it easier to generate candidate designs based on the engineering design specifications.

Table 7: Research on potential mechanisms of interest for specific locations of the robotic arm

Mechanism	General Location	Details
Brushless DC Motor - An electric motor that converts electrical energy into rotational motion without using brushes. - Electronic controllers switch current to rotate the magnetic field. - Smooth, quiet, and	Wrist Joint(s)	- Fluid / smooth wrist motion - Low torque requirements - Reduces vibration / jerk of the camera ensuring quality footage - No commutator contacts meaning consistent torque output - Allows for precise and

efficient due to no mechanical friction.		fine movement
Stepper Motors - A motor that moves in fixed angular steps instead of continuous rotation. - Provides accurate, repeatable positioning without needing feedback sensors.	Base & Arm Joints	- Stepper motors move in fixed angular steps, allowing accurate control of joint angles without needing complex feedback systems. - Easy to drive with low-cost controllers, making them ideal for hobbyist-friendly robotic arms. - Can maintain position when powered, which helps the arm hold steady camera poses without extra brakes or sensors.
Belt Drive - Transmits torque and motion from a motor to a joint. - Allows motors to be placed away from the moving link, reducing weight on the arm itself.	Connecting base motor to a gear in the arm joint Base Motor (input) → Arm Joint (output)	- Motor mounted near the base spins a gear/pulley attached to its shaft. - A timing belt connects this pulley to a second pulley on the joint shaft. - As the motor turns, the belt transmits rotational motion to the joint while keeping backlash low. - Ratio of the input gear/pulley to the output gear/pulley can increase torque or reduce speed through the belt drive.
Planetary Gearbox - A set of gears that reduces the motor's speed while increasing output torque at the joint. [1]	Base Joint(s)	- Delivers large torque in a compact housing which is ideal for supporting the whole arm's weight. - Perfect for the base or shoulder joints, which need to lift or rotate the heaviest parts of the arm. - Fits neatly inline with the motor, saving space in the

		base.
Scotch And Yoke	To make the base or arm foldable (May not be used)	- Converts rotational motion into linear motion - Rotation disk with a pin that connects to a slot moving a shaft in a linear way
Rack and Pinon		- Converts rotational motion into linear motion - Pinion spins to move a rack linearly

Appendix C: Rough drawings and extended descriptions of Candidate Design 1

This appendix section outlines the detailed technical descriptions of Candidate Design 1 and the iterations made on it.

Candidate Design #1 Description:

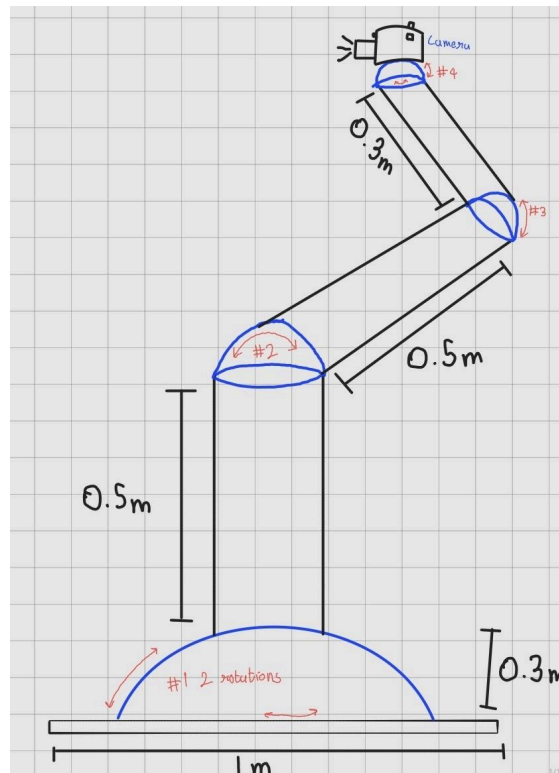


Figure 17: Visual Sketch of Candidate Design #1

- Total Of 4 joints, but 2 of the 6 joints can rotate in 2 different axis
- Joints #1 and #4 allow for movement in 2 axis.
- #1 is the base allowing 2 forms of rotation to spin in place as well as adjust height
- #4 allows 2 forms of rotation to allow stabilization and different angle shots of the camera
- The rotation of the base parallel to the ground will be controlled at the lowest point, and the rotation moving up and down will be inside the dome like structure, so the mechanism will spin along with the base.
- For join 4 there will be a way to hold the camera, this will rotate up and down allowing for angled shots. The joint connecting this end effector will spin allowing for rotation in a different axis
- Joints #1 #2 #3 will have Stepper Motors and Joint #4 will have Brushless DC Motors.
- Join #1 will have a planetary gear box maximizing torque as this will have most of the weight
- Join #4 will carry low weight and must be incredibly precise with smooth and fast movements and this will have a low torque gear box

Iterations:

- 1) Ball joints are very hard to make work with this design, so it is better to change it into multiple joints per degree of freedom
- 2) The base should have a planetary gear box at the bottom to rotate about Z axis, above that can be a separate joint that allows rotation about a different plane.

Appendix D: Rough drawings and extended descriptions of Candidate Design 2

This appendix section outlines the detailed technical descriptions of Candidate Design 2 and the iterations made on it.

Candidate Design #2 Description:

- The robotic arm has a total of six joints, labelled 1 through 6 in the diagram.
- Joints 2, 3, and 4 allow movement in the same plane, which is primarily the vertical plane of the arm.
- Joints 1 and 5 allow rotation in a different plane, aligned horizontally and perpendicular to the plane of joints 2, 3, and 4.
- Joint 6 allows rotation that is perpendicular to all other joints, providing full 3D orientation for the end-effector.
- Joints 2 and 3 are mechanically connected with a belt drive, allowing a single motor at Joint 2 to drive motion in Joint 3.
- Joints 1 and 2 use stepper motors for high torque and zero backlash, critical for precise positioning at the base and shoulder of the arm.

- Joints 4, 5, and 6 are driven by brushless DC motors, providing smooth and accurate motion for the upper arm, wrist, and end-effector.
- Joint 2 incorporates a planetary gearbox to increase torque, as it supports the majority of the arm's weight.
- The base width of the arm is 0.30 m, providing stability for the vertical movements.
- A camera is mounted near Joint 6, allowing flexible orientation and clear viewing for tasks requiring visual feedback.

Iterations:

1. Add a linear actuator that will allow for extended reach for the robot arm. This will increase usability allowing for more advanced shots and more control over close up shots where the camera can zoom in slowly.
2. Adjust belt drive mechanism such that the motor driving the belt drive which controls Joint 3 is not the same as the motor driving joint 2. This allows for more controlled motion if the linkage from joint 3 to 4 needs to move independently of the linkage between joints 2 and 3.
3. Addition of a planetary gearbox for Joint 1 allowing a higher torque for the rotation of the entire arm. This will be necessary to ensure the arm and the payload is well supported and can move without failure.

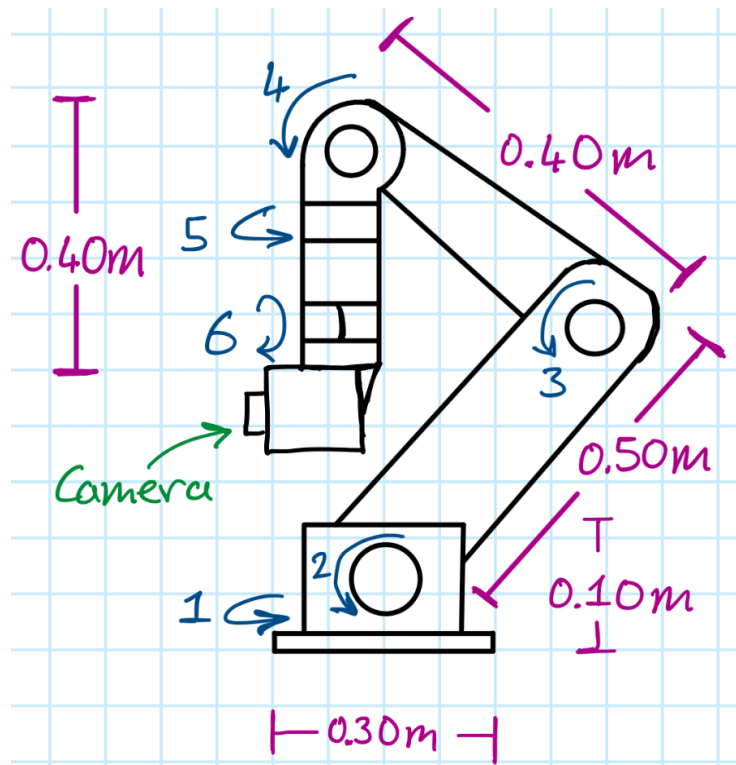


Figure 18: Visual Sketch of Candidate Design #2

Appendix E: Rough drawings and extended descriptions of Candidate Design 3

This appendix section outlines the detailed technical descriptions of Candidate Design 3 and the iterations made on it.

Candidate Design #3 Description:

- The design is based of a Camera Jib and has a total of 5 Joints
- This candidate design focuses on focuses on affordability
- Joint #1 will use a **stepper motor** with a **planetary gearbox** for large amounts of torque, as it is supporting the weight of the entire design.
- Joint #2 will use a **DC brushless motor** with a **Belt drive**, to provide torque and precision while moving most of the motor assembly to the base instead of the top, which increases stability and reduces vibrations.
- Joint #3 and #4 will utilize a **rack and pinion system** for its motorized extension and retraction for better control.
- Joint #5 will use a **DC brushless motor** for a smooth a fluid way to make sure the Gimbal Attachment point stays level
- This design by itself does not achieve the 6 axis movement, but with the Gimbal Attachment point, the user can attach any existing gimbal, which they are extremely likely to own, to the end of the arm to achieve rotational stabilization and movement.
- Due to the sandbags or weights at the other end of the camera arm, the whole arm will be balanced, and therefore Joint #2 doesn't need a large amount of torque to move and keep the arm in its required place. Therefore, cheaper and more accessible motors can be used to power the arm
- While not a conventional Robotic Arm, this candidate design offers a different design that focuses more on reach, affordability, and practicality, as this design can be used a regular Camera Jib when needed, and doesn't offer extra unneeded functionality which gimbals can already achieve

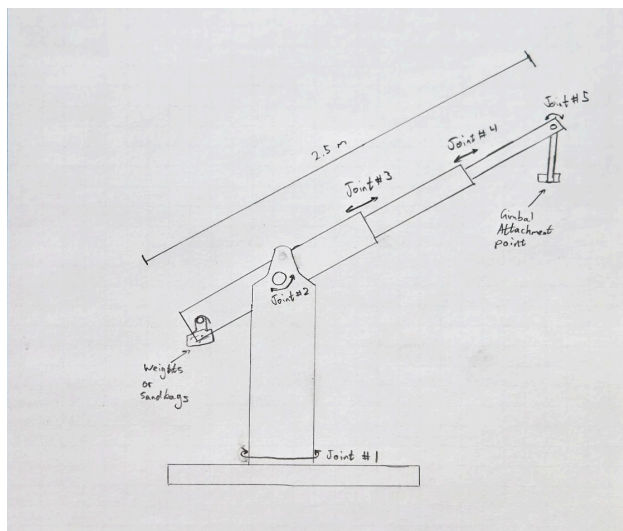


Figure 19:

Appendix F: Rough drawings and extended descriptions of Candidate Design 4

This appendix section outlines the detailed technical descriptions of Candidate Design 4 and the iterations made on it.

Candidate Design #4 Description:

- Total joints = 7
- The design also uses extension/retraction mechanisms like design 3
- Joint #1 will use a **multi-stage telescopic actuator**, where several tubes of decreasing diameter slide out from one another. The drive motor is mounted in the largest, innermost tube, and motion is transmitted to each tube through a **Dual-Belt Closed-Loop System** – one belt pulls the stages out and another belt acts as a tension stabilizer.
- Joint #4, #5, and #6 (Camera Yaw, Pitch, Roll): These three axes are integrated into a pre-existing, off-the-shelf 3-Axis Gimbal or Camera Head.
- Joint #7 is for **Ground Bracing and Stability**: The strut can be extended to make contact with the ground, a table, or a prop.
- This design is optimal for extended reach and stability

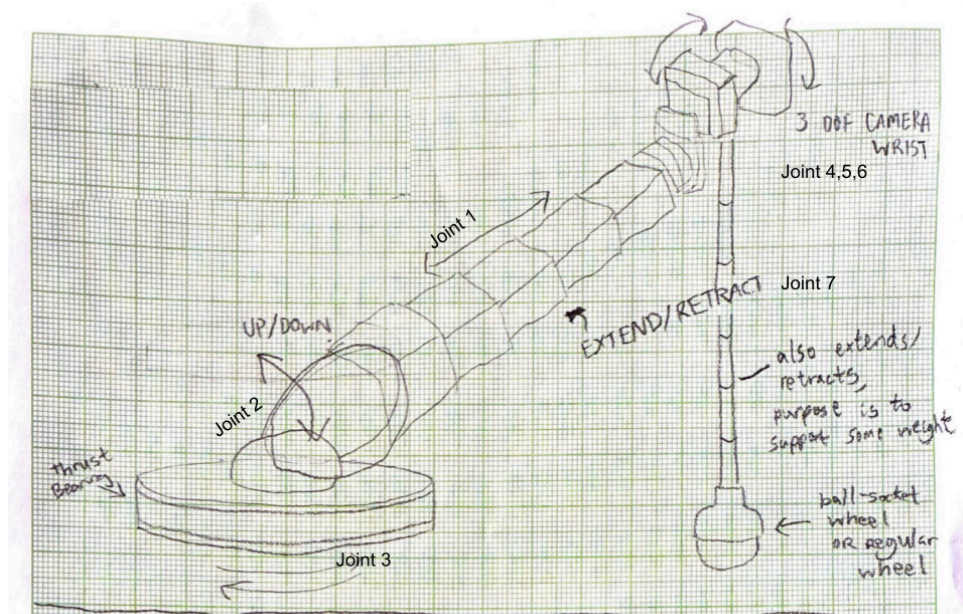


Figure 20: Visual Sketch of Candidate Design #4

Appendix G: Research for parts and motors for the final CAD design

This appendix describes

Table 8: Final design components research

PART:	DESCRIPTION:
Lead Screw [12]	- Lead screw for the linear actuator - Quantity: 2
QSH8618-96-55-700 [13]	- Strong stepper motor - \$420.61 - 7Nm torque - (Used for arm rotation on base)
UltraPlanetary Cartridge - 5:1 [14]	- 5:1 gearbox (scalable) - Quantity: 2
UltraPlanetary Cartridge - 3:1 [15]	- 3:1 gearbox - Quantity 2
UltraPlanetary Female 5mm Hex Output V2 [16]	- Output for the gearbox
NEO Vortex Brushless Motor [17]	- NEO motor - 3.6 N M torque - 6784 RPM (probably for 2nd part of arm or to rotate base)
High Torque Timing Belt [18] Tooth Timing Pulley [19]	- HTD 8M timing belt (belt needs to be double the length based on your design) - HTD 8M pulley aluminum, 20 teeth
SparkFun Electronics 16413 [20]	- Servomotor - \$12.69 - Seems to be more for toys - Only servomotor under \$500 on digikey with a CAD model
DC Servomotor with Integrated Drive [21]	- DC Servomotor - \$427.69 - 0.4067 Nm torque
Lin Engineering WO-4418L-36 [22]	- Stepper Motor
Ball Bearing [23]	- 100mm ball bearing

Ball [24]	- 80mm ball bearing
-----------	---------------------

Robotic Arm Guide [25]

- Very basic start for robotic arm design
- Think of actuators as muscles and controllers as brain (we do not need to worry about controllers as this is a mechanical course)
- “Articulated arms are the most common design, with rotary joints that give them human-like flexibility for welding, painting, and assembly.” This is likely going to be a huge part of our design, a lot of rotary joints.

Component	Role
Mechanical structure	Links and joints form the skeleton. Gives the arm strength and flexibility to rotate. Bend and extend like a human arm
Actuators and motors	Work as muscles. Converts electric, hydraulic, or pneumatic power into motion. Electric motors dominate modern arms for precision
Controller	The brain that processes commands and coordinates all joints and actuators. Supports both simple routines and advanced AI control
Sensors	Provide feedback on position, torque, and speed. Enables real-time corrections for accuracy and safety
End effector	The tool at the wrist that performs gripping, welding, painting, or surgery. Swappable for different applications

-

- Take the table above into consideration when designing

Overall all this website was very helpful for getting started, a lot of the information was regarding electrical and programming aspects.

Different types of arms [26]:

Articulated arm: 5 or more joints or degrees of freedom, this means ours is an articulated arm

Six-axis: is a type of articulated arm with 6 degrees of freedom. This is the most common type nowadays

Collaborative robot: meant to be user friendly as they stop before hitting something, this makes them super safe to work around.

SCARA: less flexible but they are specified to do certain tasks

Cartesian: moves with an XYZ coordinate system

Cylindrical: up down and rotational movement

Parallel/Delta: used in factory lines, they have a limited movement. Very very fast.

Bearings are necessary to reduce friction, have them anywhere where things are rubbing against each other [27].

Linkages can be useful for rotational motion, using a bar or rod to connect 2 pieces together, which gives it freedom to rotate [27].

“Revolute joints, also known as rotary joints, facilitate rotation around a single axis. They offer versatility in movement and are pivotal in enabling bending and twisting motions [27].” will be most of our robot since we rely heavily on revolutions.

Ball and socket joints are harder to pull off, so we will be avoiding this.

Prismatic joints are used for linear motion, in the case of a linear actuator.

There are a lot of types of electrical linear actuators, these are really good for fine precision but they are expensive and oftentimes limited for range of motion [28].

Hydraulic is also an option but then we need to worry about a hydraulic system with an air compressor and all. This could be challenging.

Ball and lead screws are simple, rotational to linear motion conversion [29].

Ball screws are smoother but they have lower stall torque and they require lubrication [29].

For our case a lead screw is better, it does not need to be incredibly smooth but there is a requirement for high stall torque, or else it would fall back down.

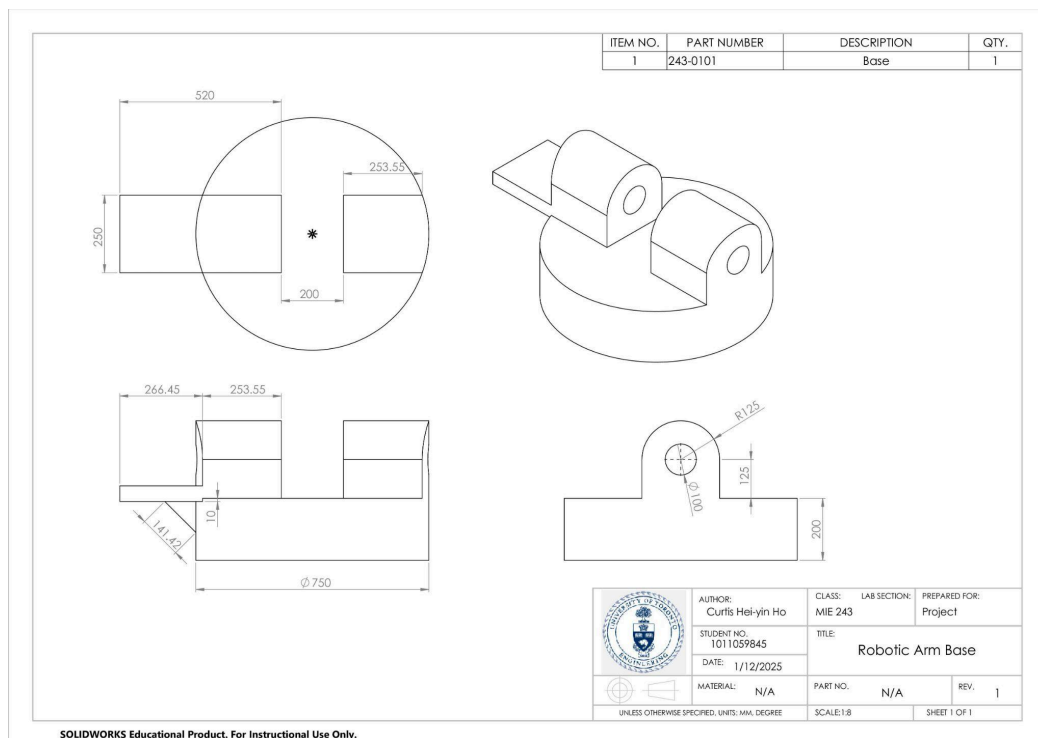
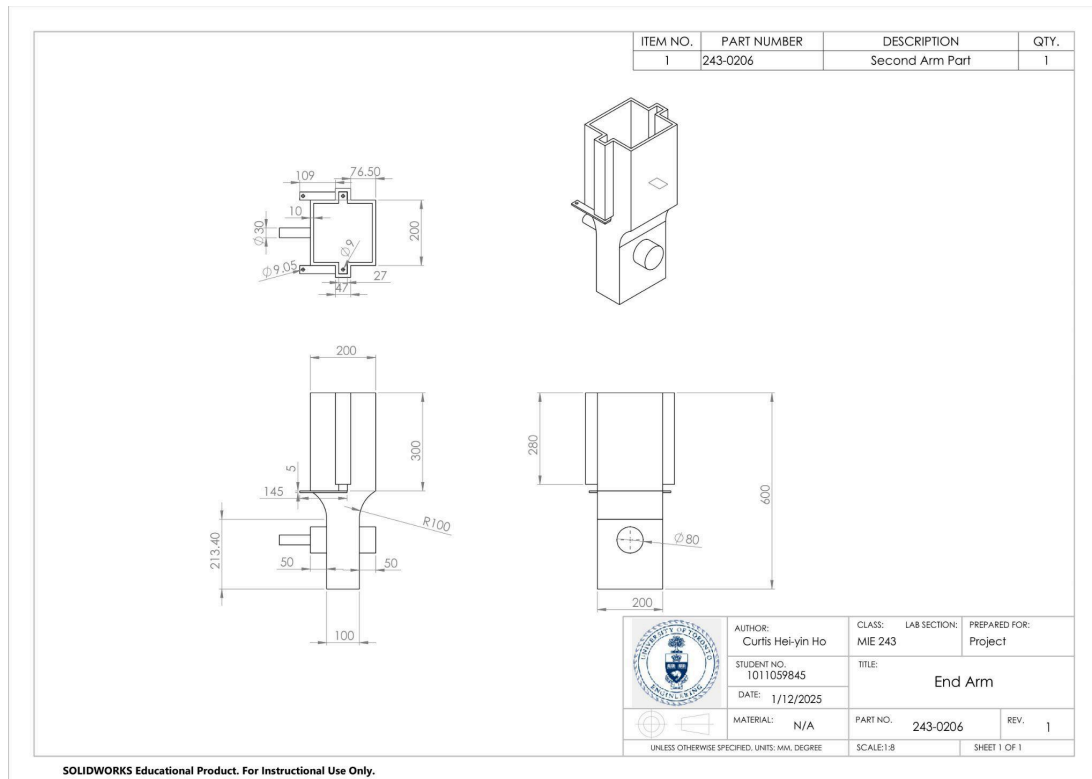
Servo: sacrifices torque and speed for incredible precision, very weak but VERY precise [1].

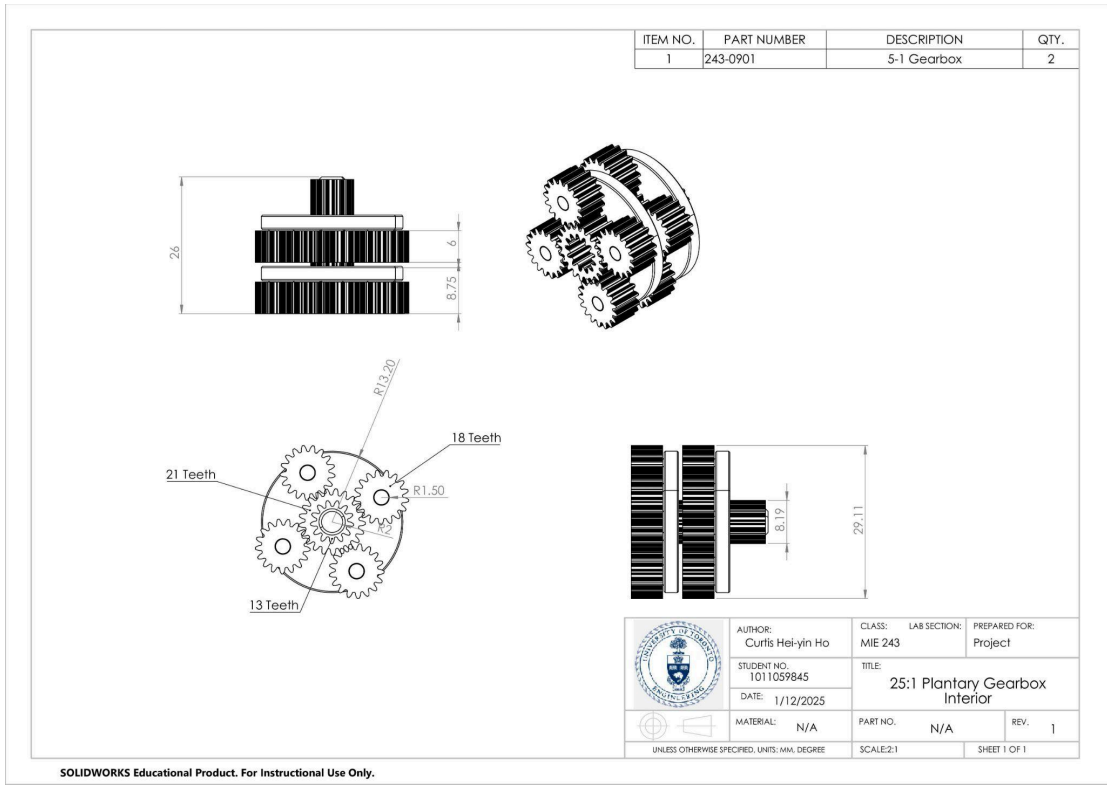
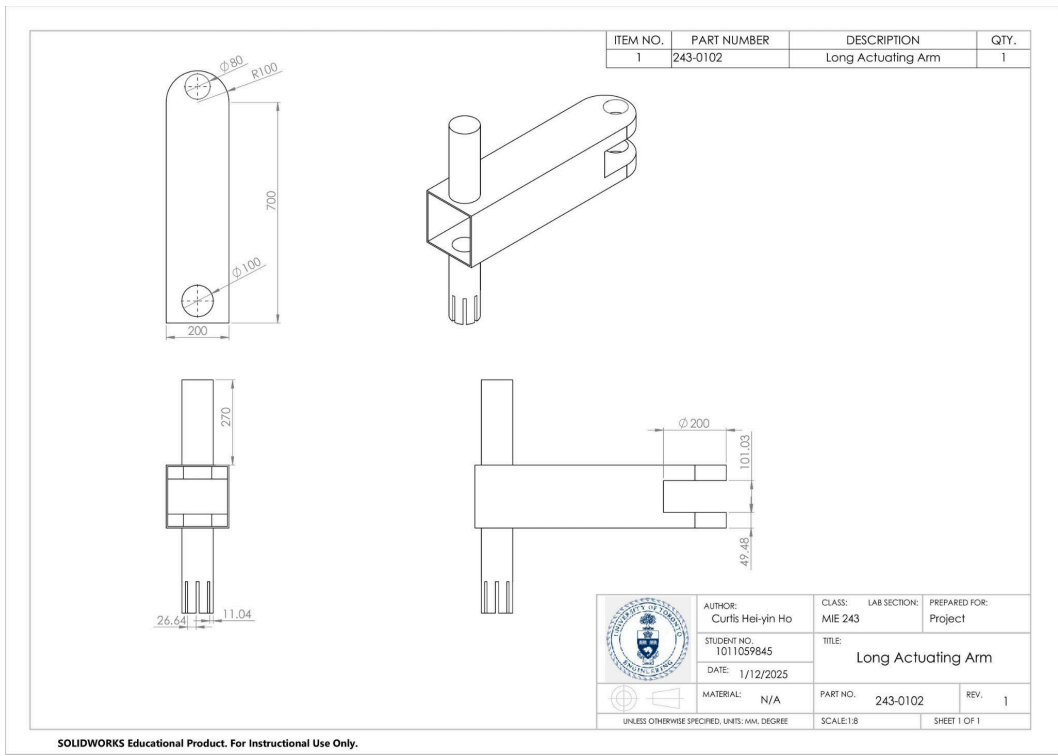
Oftentimes it is limited to 0-180 degrees of rotation, good for steering and stuff [1].

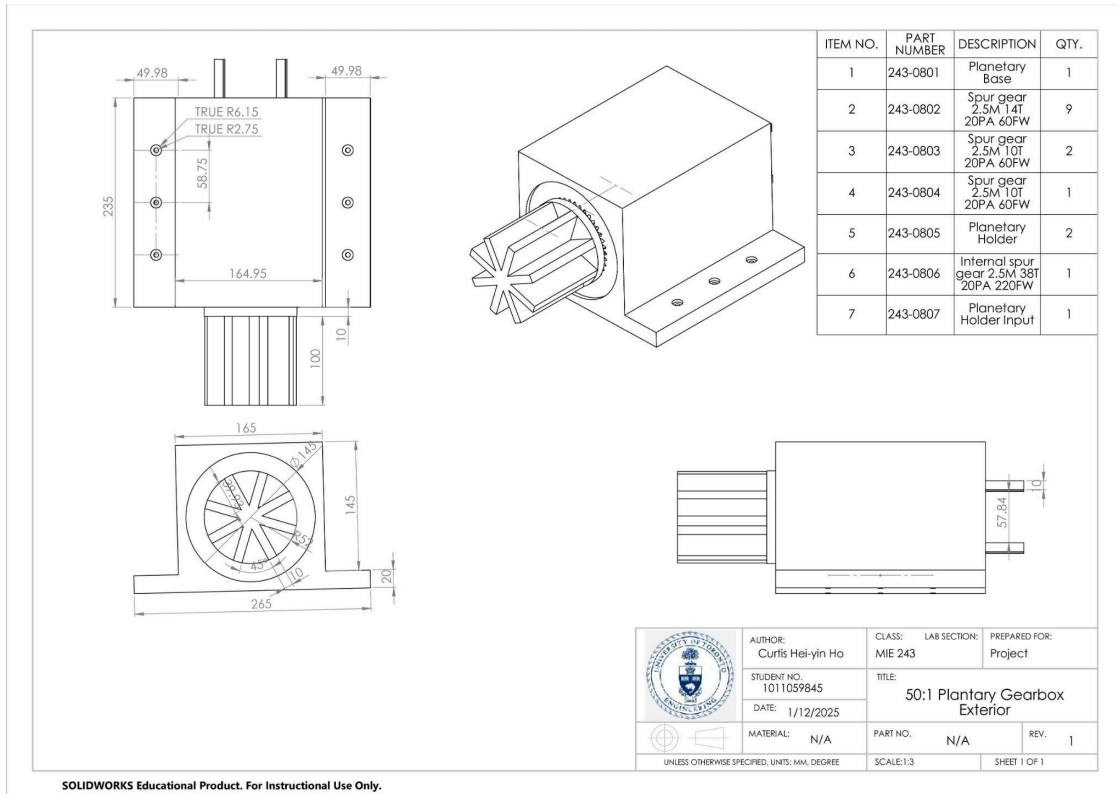
Stepper: They have discrete steps they take [1]. Set “steps” they go through this helps with repeatability and precision. They also have incredible stall torque compared to DC motors [1].

DC Motors: the most basic motor, simply just spins at a specific torque and speed [1].

Appendix G: Component Engineering Drawings







Appendix I: References

- [1] M. A. Hossain, “Planetary Gearbox: How It Works, Types, and Advantages?,” Assun, Accessed: Dec 2, 2025. [Online]. Available: <https://assunmotor.com/blog/planetary-gearbox/>
- [2] “What are Brushless DC Motors,” Renesas. Accessed: Dec 2, 2025. [Online]. Available: https://www.renesas.com/en/support/engineer-school/brushless-dc-motor-01-overview?srltid=AfmBOop5lwtEpg5qRtj_cEEe5UAOZndgdXfi9oqC_0ushkwBrApZjsRu
- [3] S. Sild, Belt Drives & Types of Belts, Fractory. Accessed: Dec 2, 2025. [Online]. Available: <https://fractory.com/belt-drives/>
- [4] “Timing Pulleys,” Torque Transmission. Accessed: Dec 2, 2025. [Online]. Available: <https://www.torque-trans.com/timing-pulleys>
- [5] S. Sild, Lead Screws Explained, Fractory. Accessed: Dec 2, 2025. [Online]. Available: <https://fractory.com/lead-screws/>
- [6] “What's the Difference Between DC, Servo & Stepper Motors?” The Pi Hut. Accessed: Dec 2, 2025. [Online]. Available: <https://thepihut.com/blogs/raspberry-pi-tutorials/whats-the-difference-between-dc-servo-and-mp-stepper-motors?srltid=AfmBOoqmoCuEg2k2UkW6vI1woIUvrUN1ZFF9YqL9ieTfqeQi-YT120Tb>
- [7] C. E. Selector, Aluminum Specifications, University of Cambridge, 2014.
- [8] AFIMAC Canada, “Factory Worker/General Labour,” ZipRecruiter. Accessed: Dec 2, 2025. [Online]. Available: <https://www.ziprecruiter.com/Jobs/Factory-Worker/-in-Toronto,ON?version=next&lvk=8Y3iC20bC75fKGbNMTIIPg.--O3SjzODpF>
- [9] Nikon Company, “Bolt™,” MRMC, Surrey, United Kingdom. Accessed: Dec 2, 2025. [Online]. Available: https://www.mrmoco.com/wp-content/uploads/dlm_uploads/2025/04/BOLT-Leaflet.pdf
- [10] “Universal Robots UR3e Collaborative Robot Arm,” Vention. [Online]. Available: vention.io/parts/universal-robots-ur3e-collaborative-robot-arm-2446

- [11] “EVO The Robot That Moves With You,” Motorized Precision. [Online]. Available: <https://www.motorizedprecision.com/evo>
- [12] “Precision Lead Screws and Nuts,” McMaster-Carr. Accessed: Dec 2, 2025. [Online]. Available: <https://www.mcmaster.com/products/lead-screws/precision-lead-screws-and-nuts-1~/length~300-mm/>
- [13] “Analog Devices Inc./Maxim Integrated QSH8618-96-55-700,” Digikey. Accessed: Dec 2, 2025. [Online]. Available: <https://www.digikey.ca/en/products/detail/analog-devices-inc-maxim-integrated/QSH8618-96-55-700/4843436>
- [14] “UltraPlanetary Cartridge - 5:1,” REV Robotics. Accessed: Dec 2, 2025. [Online]. Available: https://www.revrobotics.com/rev-41-1603/?searchid=4900638&search_query=25%3A1+planetary
- [15] “UltraPlanetary Cartridge - 3:1,” REV Robotics. Accessed: Dec 2, 2025. [Online]. Available: <https://www.revrobotics.com/rev-41-1601/>
- [16] “UltraPlanetary Female 5mm Hex Output V2,” REV Robotics. Accessed: Dec 2, 2025. [Online]. Available: https://www.revrobotics.com/rev-41-1615/?searchid=4901610&search_query=
- [17] “NEO Vortex Brushless Motor,” REV Robotics. Accessed: Dec 2, 2025. [Online]. Available: <https://www.revrobotics.com/rev-21-1652/>
- [18] “480-8M-30 | High Torque Timing Belt,” FORCES. Accessed: Dec 2, 2025. [Online]. Available: <https://www.revrobotics.com/rev-21-1652/>
- [19] “Tooth Timing Pulley,” AliExpress. Accessed: Dec 2, 2025. [Online]. Available: www.aliexpress.com/p/tesla-landing/index.html?scenario=c_ppc_item_bridge&productId=1005006585380355
- [20] “SparkFun Electronics 16413,” Digikey. Accessed: Dec 2, 2025. [Online]. Available: www.digikey.ca/en/products/detail/sparkfun-electronics/16413
- [21] “DC Servomotor with Integrated Drive,” McMaster-Carr. Accessed: Dec 2, 2025. [Online]. Available: <https://www.mcmaster.com/5151N111/>

- [22] “Lin Engineering WO-4418L-36,” Digikey. Accessed: Dec 2, 2025. [Online]. Available: <https://www.digikey.ca/en/products/detail/lin-engineering/WO-4418L-36/11564514>
- [23] “Deep Groove Ball Bearing,” Temu. Accessed: Dec 2, 2025. [Online]. Available: [/www temu.com/ca/6816-2rs-deep--ball-bearing-80x100x10mm--chrome-steel-61816-2rs-bearings](https://www temu.com/ca/6816-2rs-deep--ball-bearing-80x100x10mm--chrome-steel-61816-2rs-bearings)
- [24] “Deep Ball Bearing,” Temu. Accessed: Dec 2, 2025. [Online]. Available: www temu.com/ca/6307-2rs-deep--ball-bearing--x-80mm-x-21mm-double-sealed-heavy-duty-chrome
- [25] “How do robotic arms work? Principles, components, and uses.” Standard Bots. Accessed: Dec 2, 2025. [Online]. Available: <https://standardbots.com/blog/how-do-robotic-arms-work-a-comprehensive-guide>
- [26] C. Bernier. “Robotic Arms: Different Types and When to Use Them.” HowToRobot. Accessed: Dec 2, 2025. [Online]. Available: <https://howtorobot.com/expert-insight/robotic-arms>
- [27] “Types Of Robot Joints.” Dorna Robotics. Accessed: Dec 2, 2025. [Online]. Available: <https://dorna.ai/blog/types-of-robot-joints/>
- [28] “Types of Linear Actuators: Discover Their Various Makes, Considerations and Benefits.” Industrial Quick Search. Accessed: Dec 2, 2025. [Online]. Available: <https://www.iqsdirectory.com/articles/linear-actuator/types-of-linear-actuators.html>
- [29] “Ball Screws vs Lead Screws: What is the Difference?” Thomson. Accessed: Dec 2, 2025. [Online]. Available: <https://www.thomsonlinear.com/en/about>